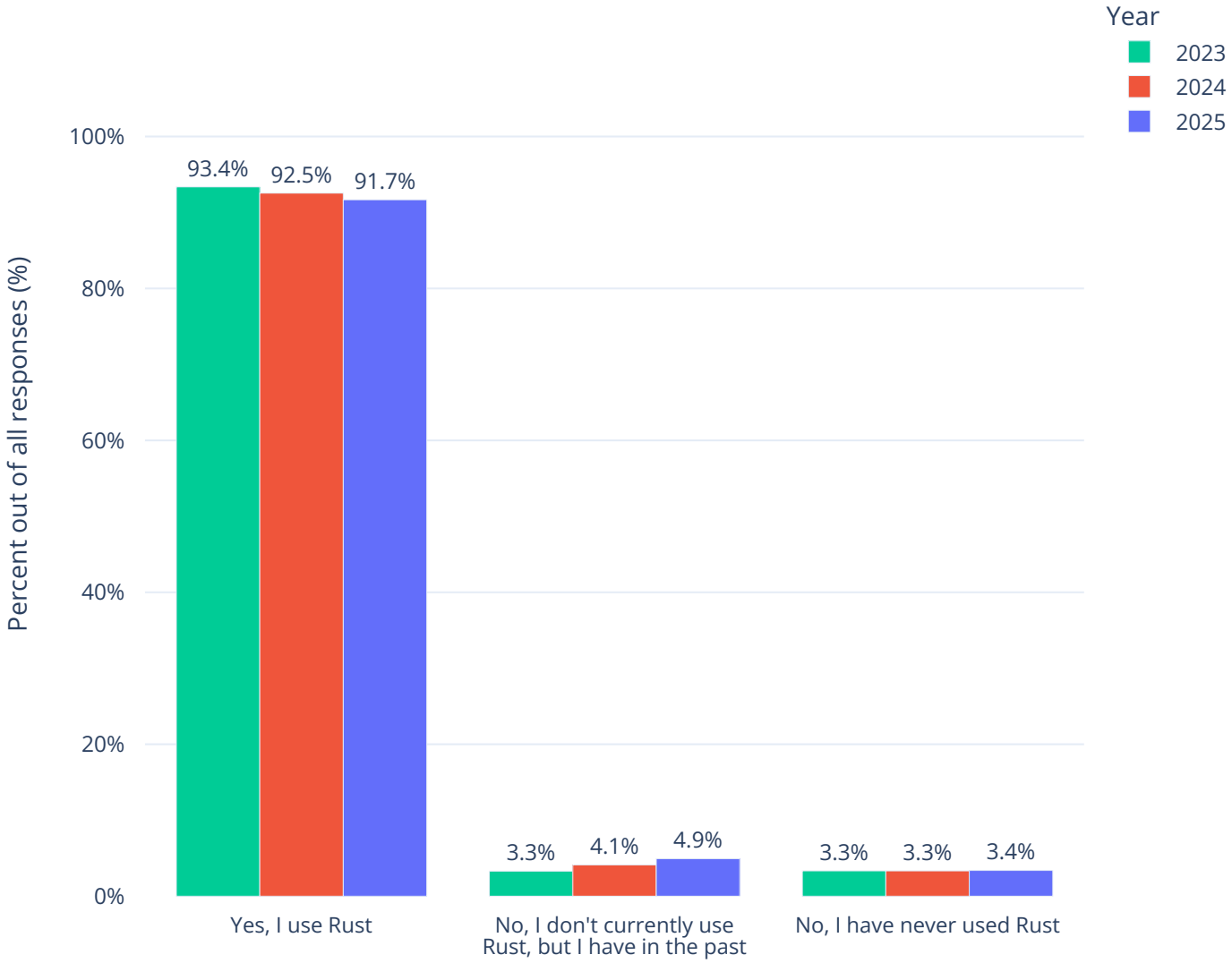
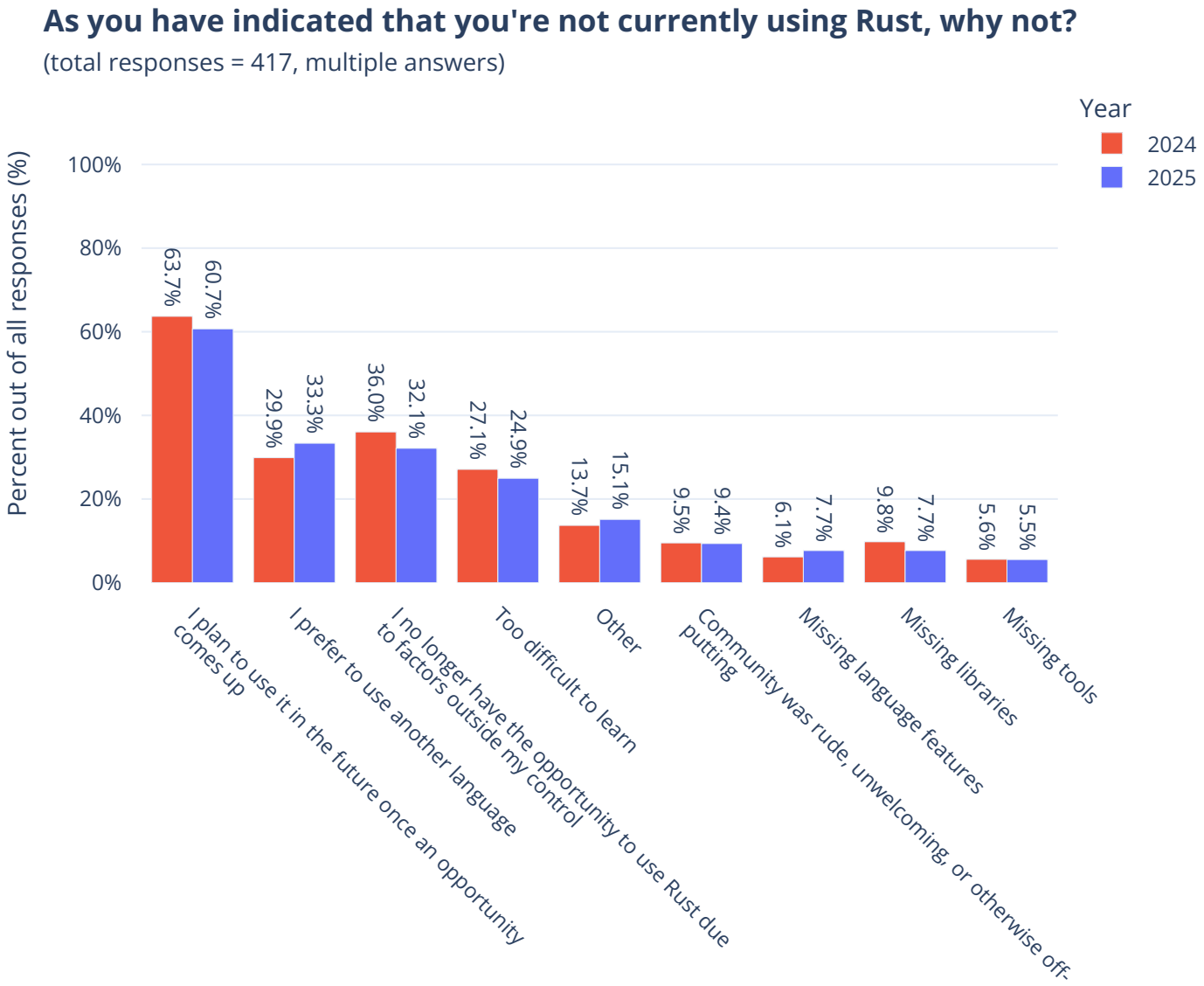
Rust Annual survey 2025 report

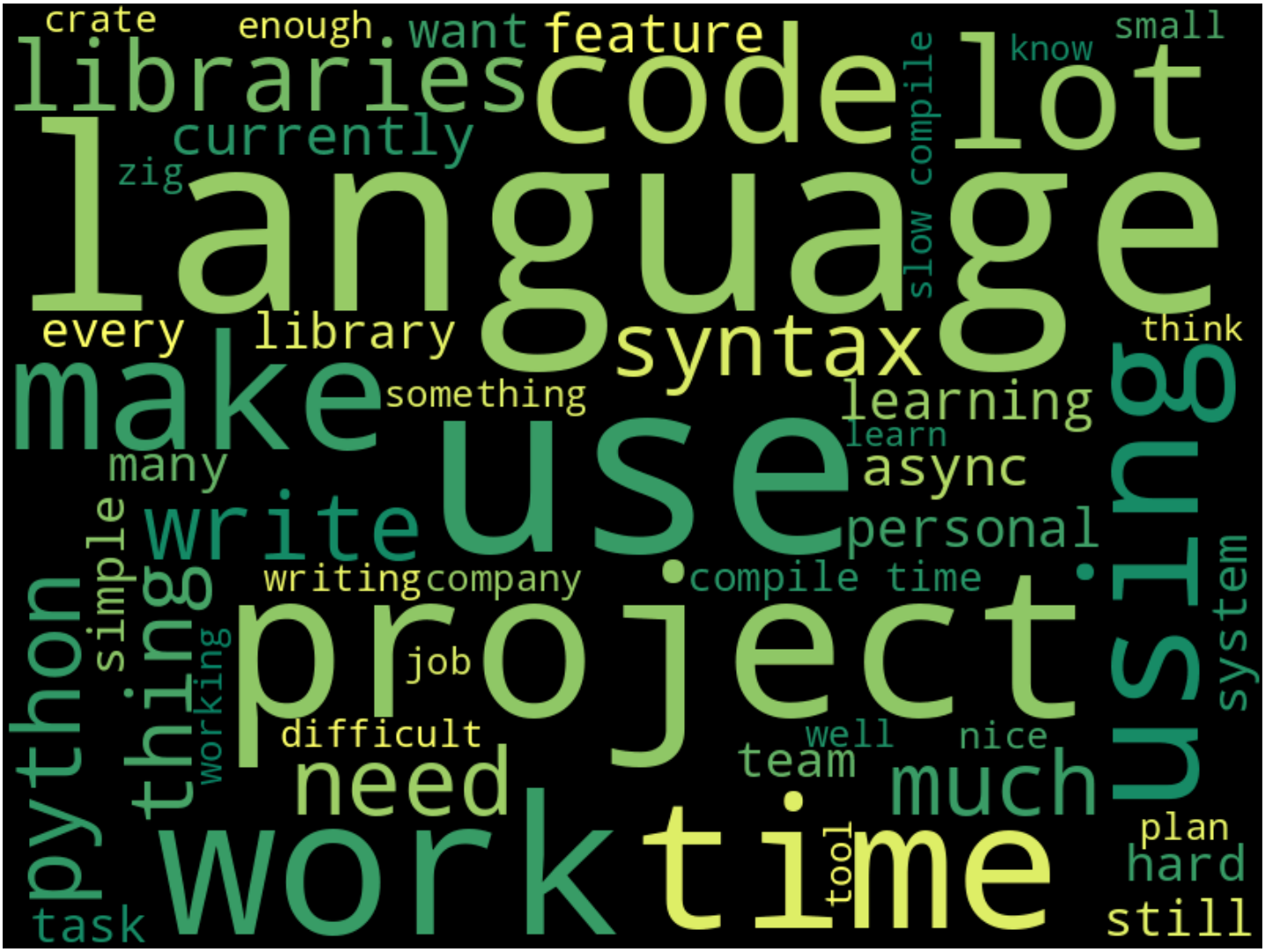
Do you use Rust?

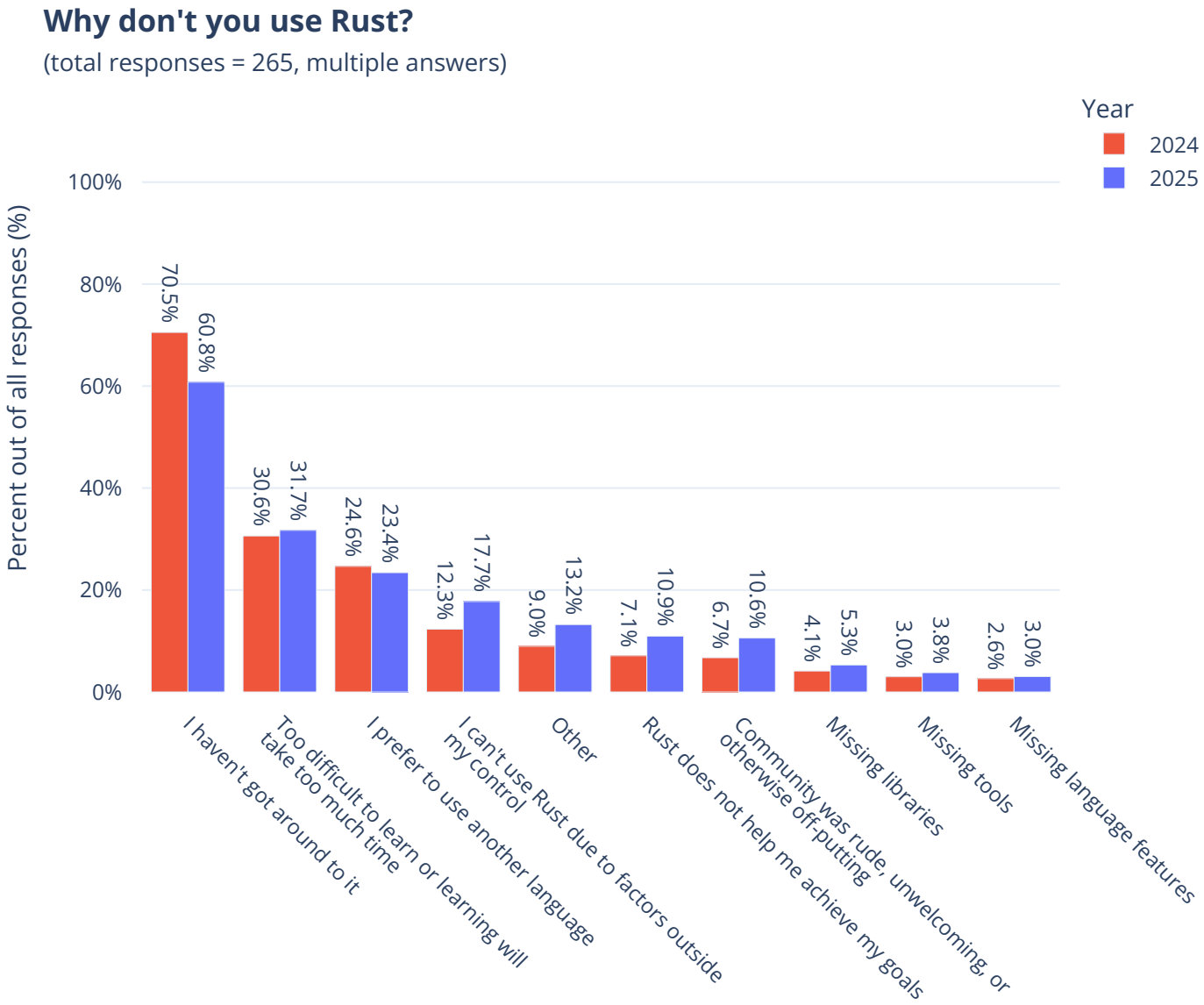
(total responses = 9354, single answer)



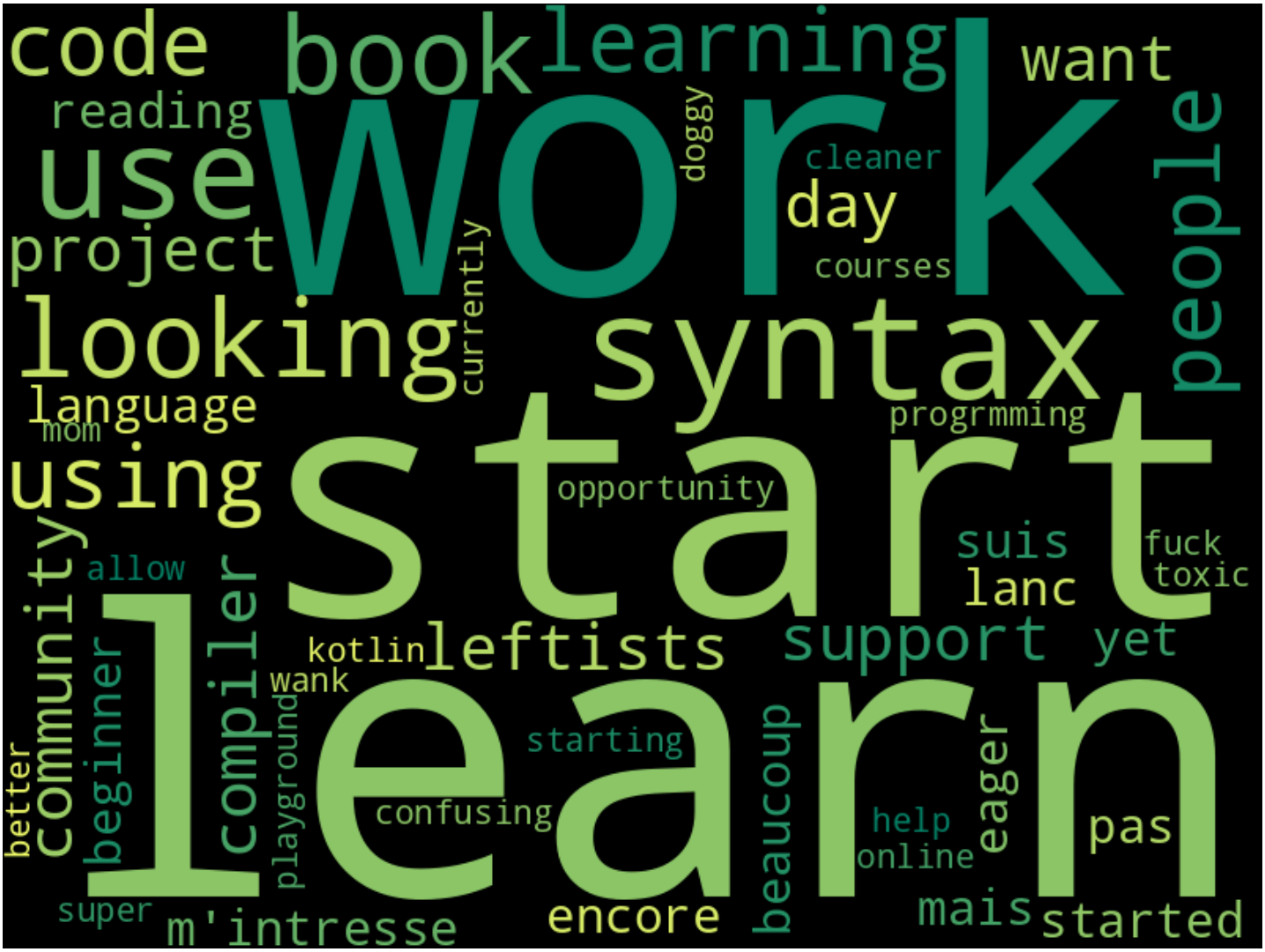

As you have indicated that you're not currently using Rust, why not?
(total responses = 417, multiple answers)
Year
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
63.7%
60.7%
29.9%
33.3%
36.0%
32.1%
27.1%
24.9%
13.7%
15.1%
9.5%
9.4%
6.1%
7.7%
9.8%
7.7%
5.6%
5.5%
I plan to use it in the future once an opportunity comes up
I prefer to use another language
I no longer have the opportunity to use Rust due to factors outside my control
Too difficult to learn
Other
Community was rude, unwelcoming, or otherwise off-putting
Missing language features
Missing libraries
Missing tools

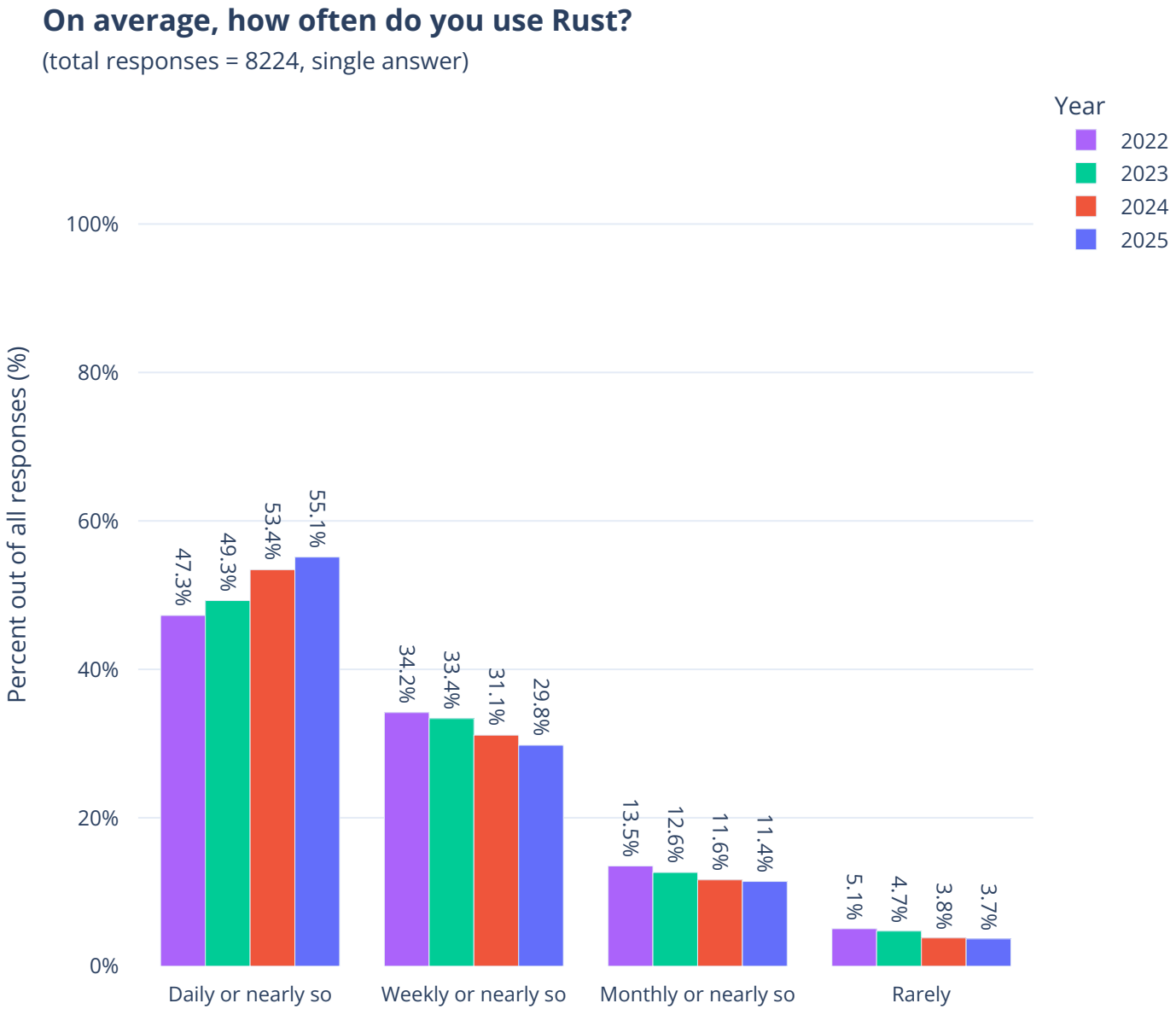
Wordcloud of open answers for the previous chart:

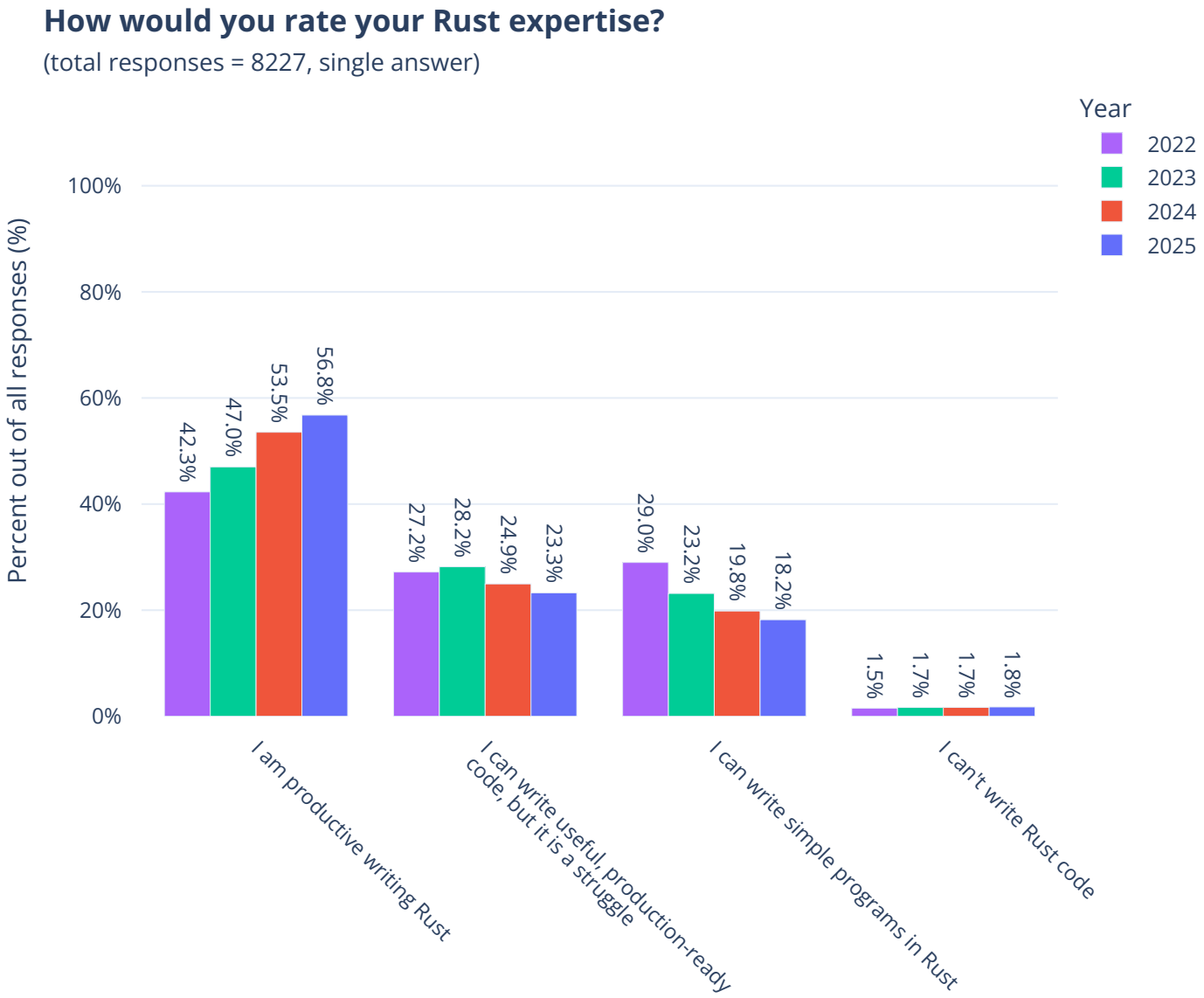



Why don't you use Rust?
(total responses = 265, multiple answers)
Year
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
70.5%
60.8%
30.6%
31.7%
24.6%
23.4%
12.3%
17.7%
9.0%
13.2%
7.1%
10.9%
6.7%
10.6%
4.1%
5.3%
3.0%
3.8%
2.6%
3.0%
I haven't got around to it
Too difficult to learn or learning will take too much time
I prefer to use another language
I can't use Rust due to factors outside my control
Other
Rust does not help me achieve my goals
Community was rude, unwelcoming, or otherwise off-putting
Missing libraries
Missing tools
Missing language features

Wordcloud of open answers for the previous chart:

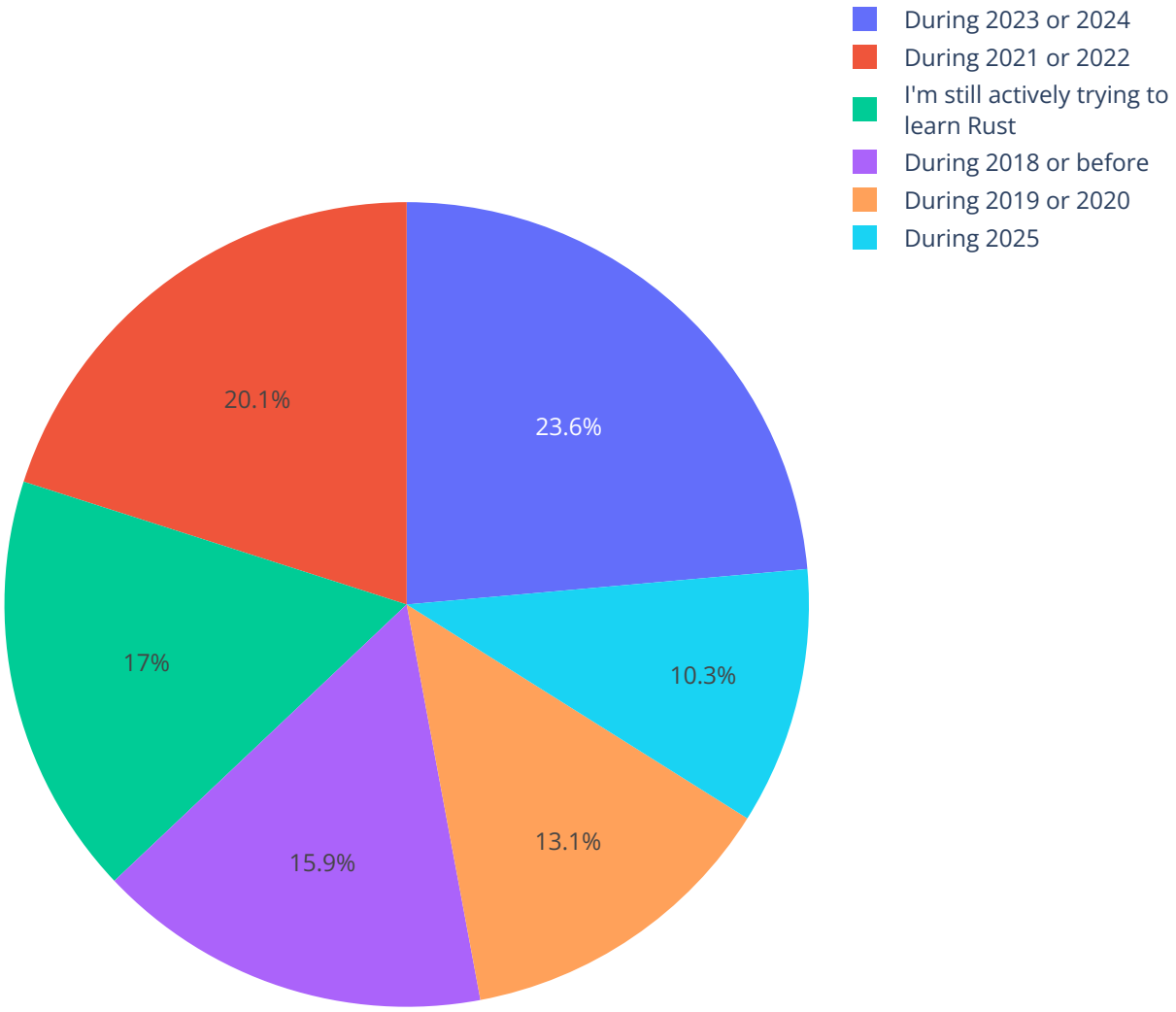



On average, how often do you use Rust?
(total responses = 8224, single answer)
Year
2022
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
47.3%
49.3%
53.4%
55.1%
34.2%
33.4%
31.1%
29.8%
13.5%
12.6%
11.6%
11.4%
5.1%
4.7%
3.8%
3.7%
Daily or nearly so
Weekly or nearly so
Monthly or nearly so
Rarely


How would you rate your Rust expertise?
(total responses = 8227, single answer)
Year
2022
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
42.3%
47.0%
53.5%
56.8%
27.2%
28.2%
24.9%
23.3%
29.0%
23.2%
19.8%
18.2%
1.5%
1.7%
1.7%
1.8%
I am productive writing Rust
I can write useful, production-ready code, but it is a struggle
I can write simple programs in Rust
I can't write Rust code

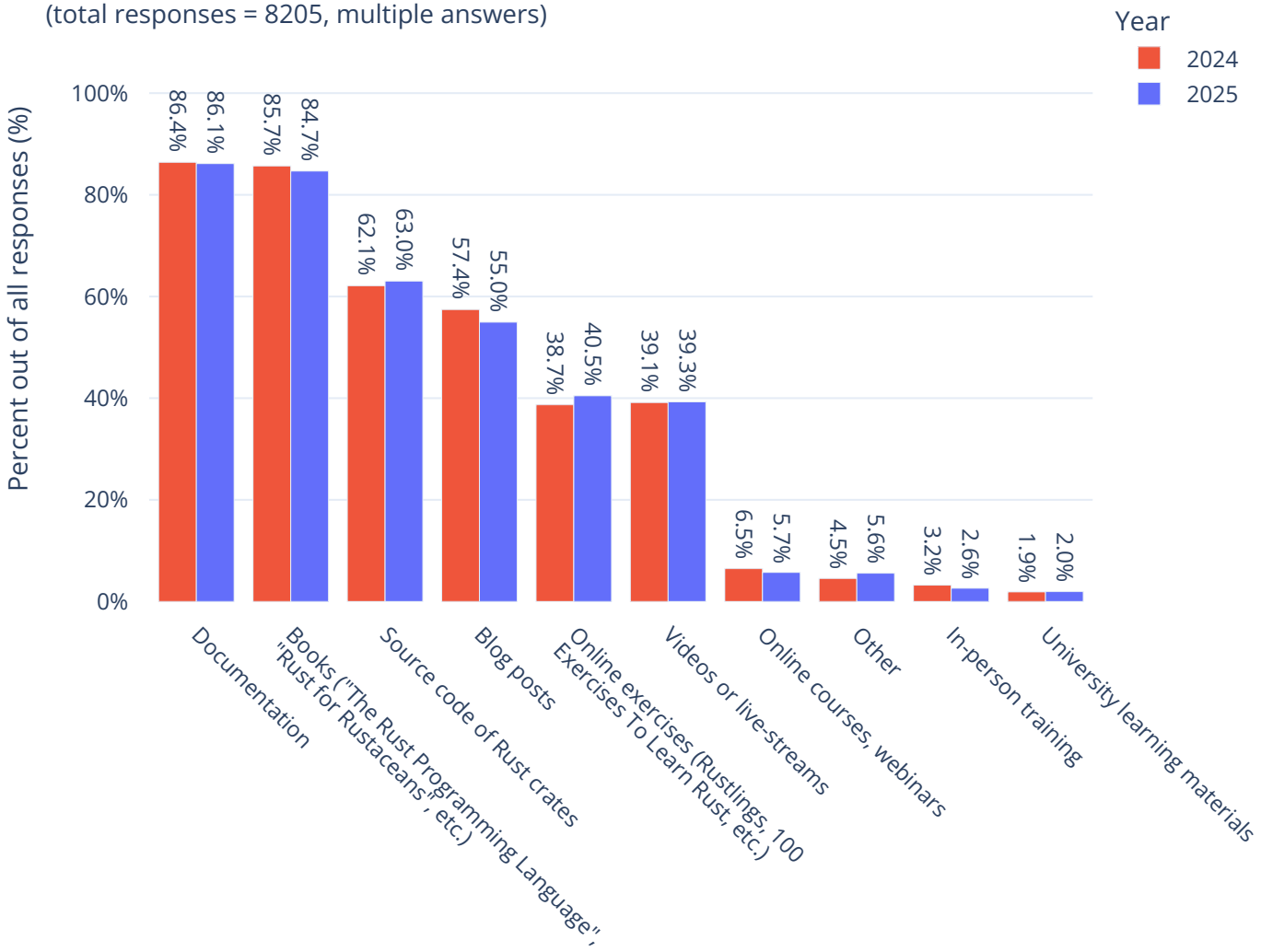
When did you learn to program in Rust?

(total responses = 8210)

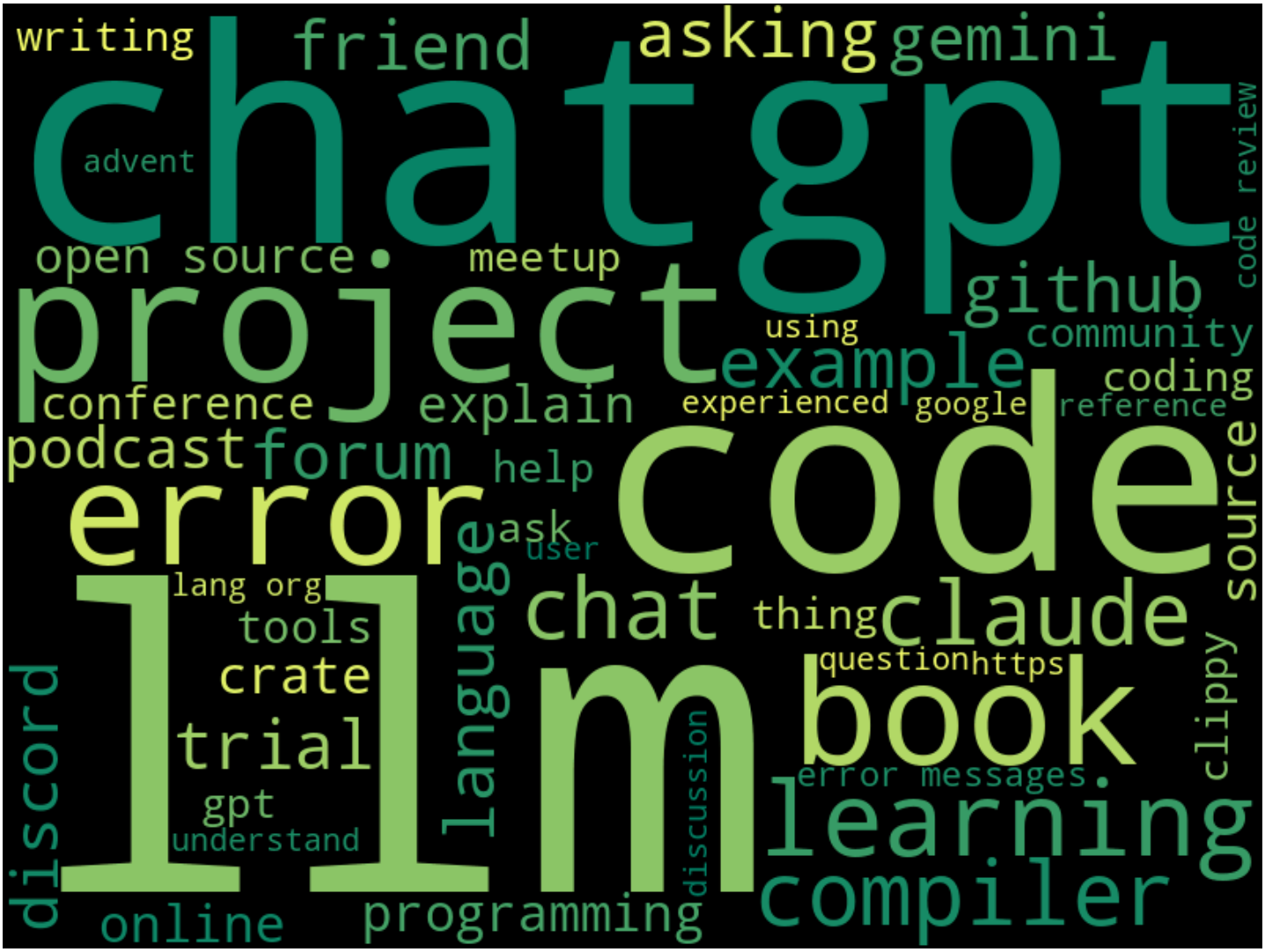


If you consumed learning material about Rust, which kind of material did you consume?

(total responses = 8205, multiple answers)

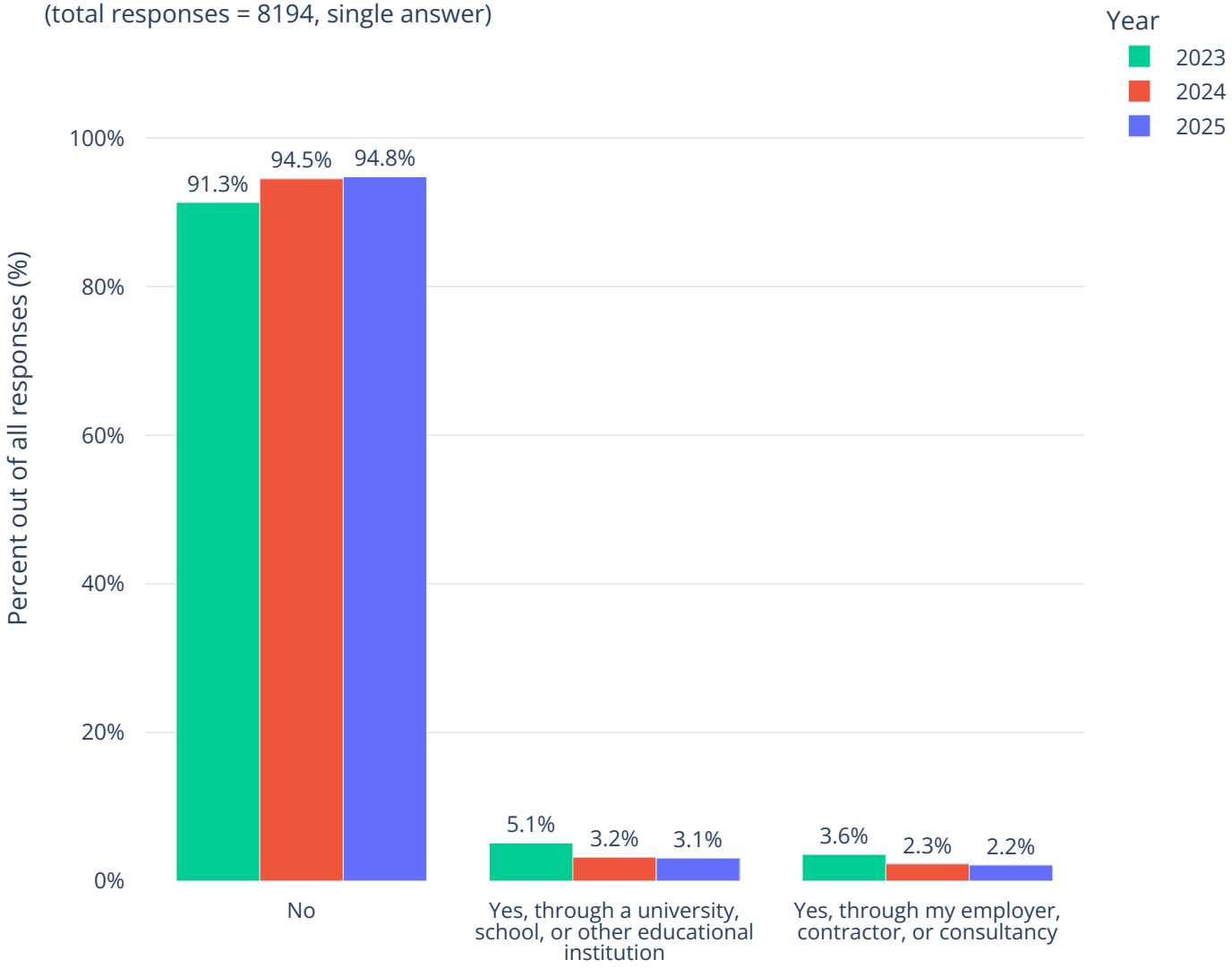


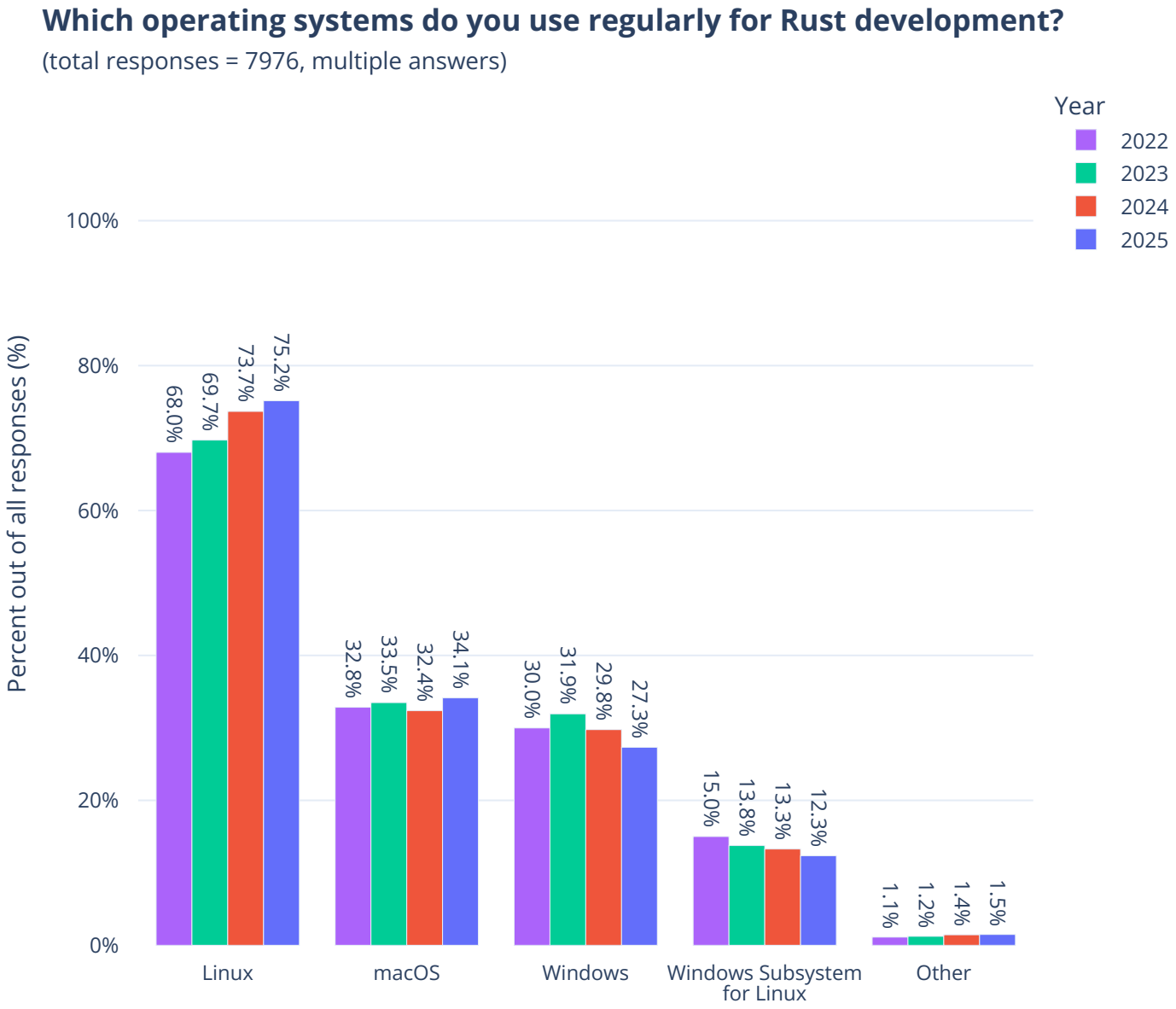
Wordcloud of open answers for the previous chart:



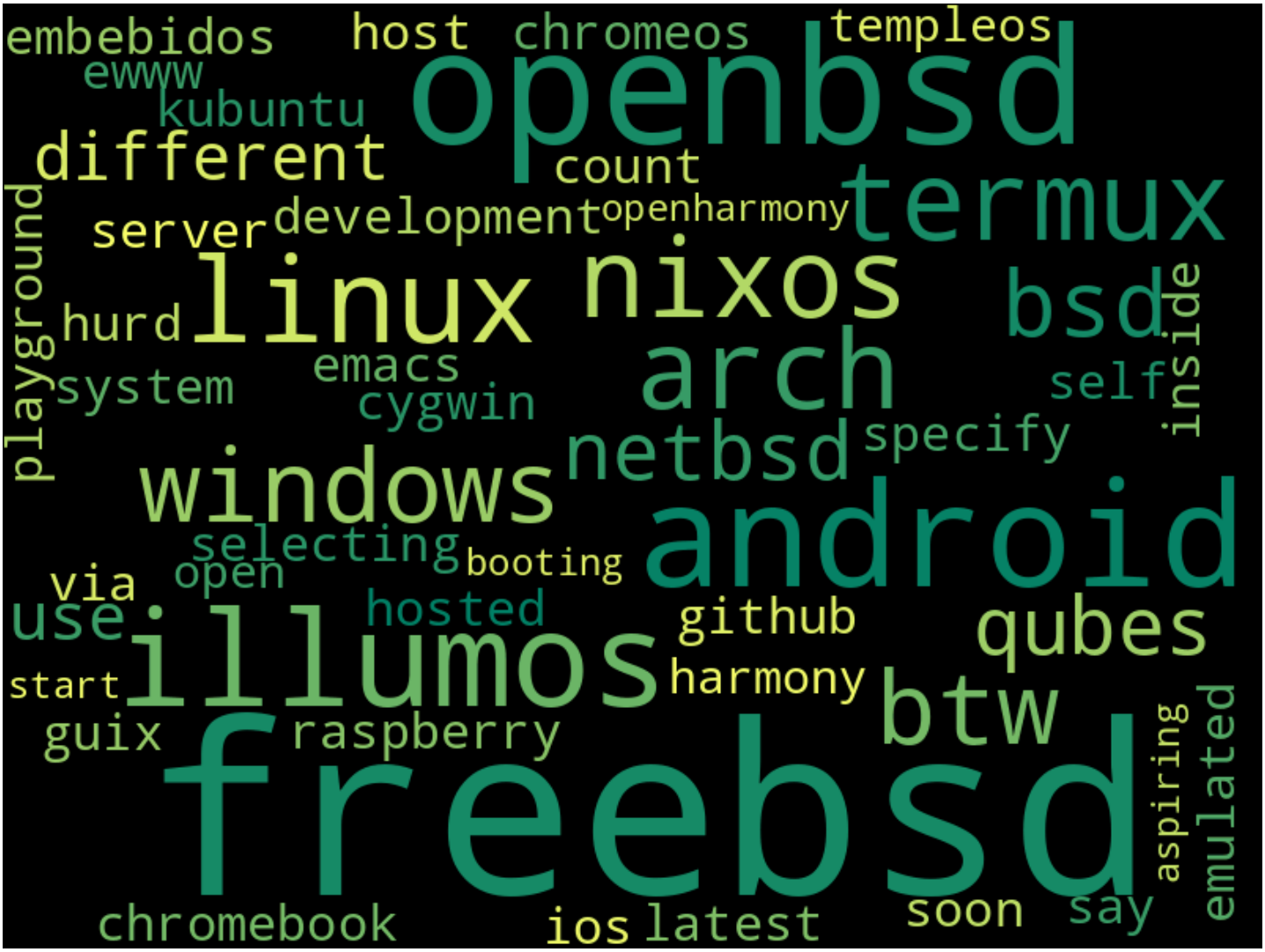
Are you currently taking a course that uses or teaches Rust OR have you taken a course of this type in the last year?

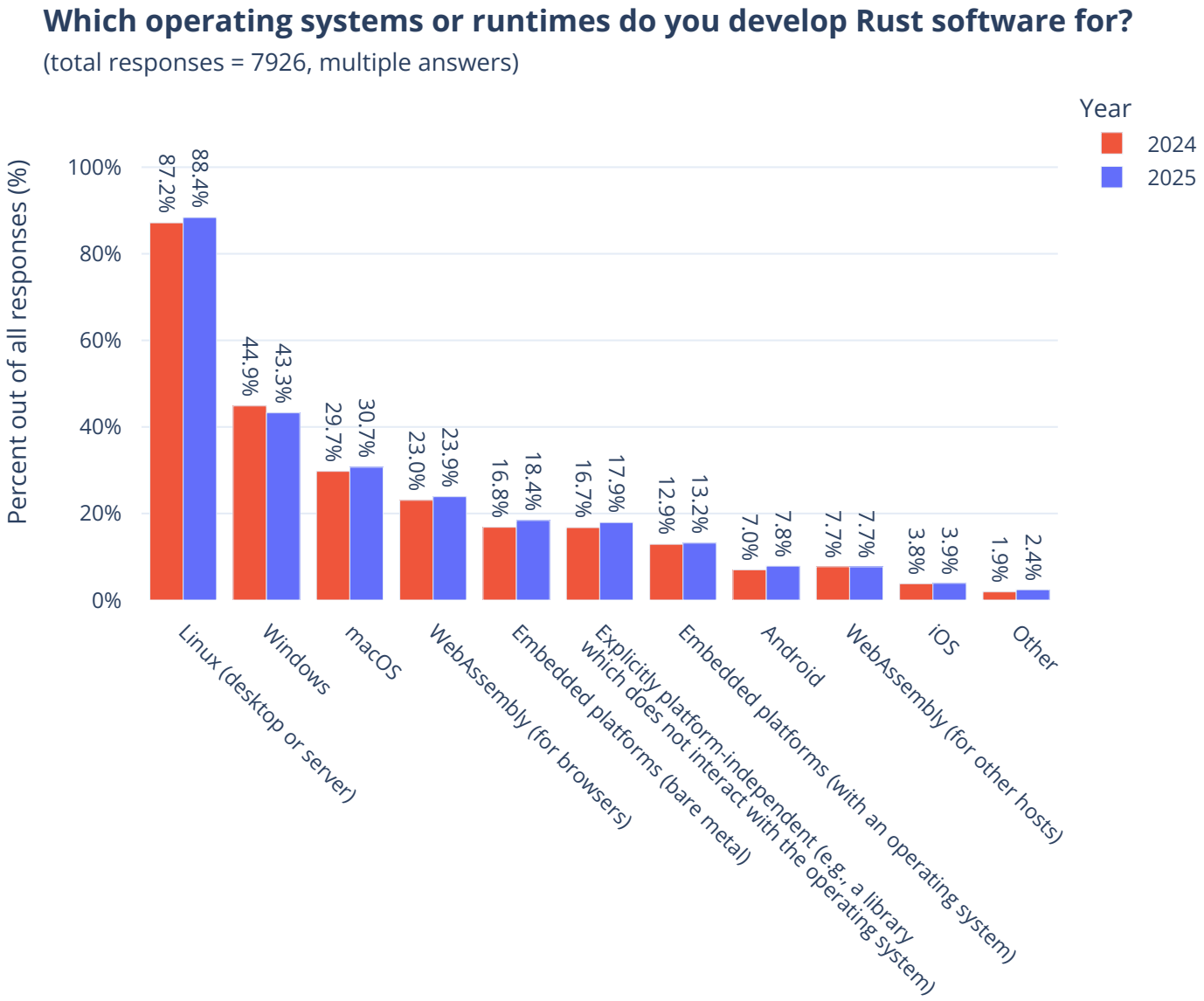
(total responses = 8194, single answer)



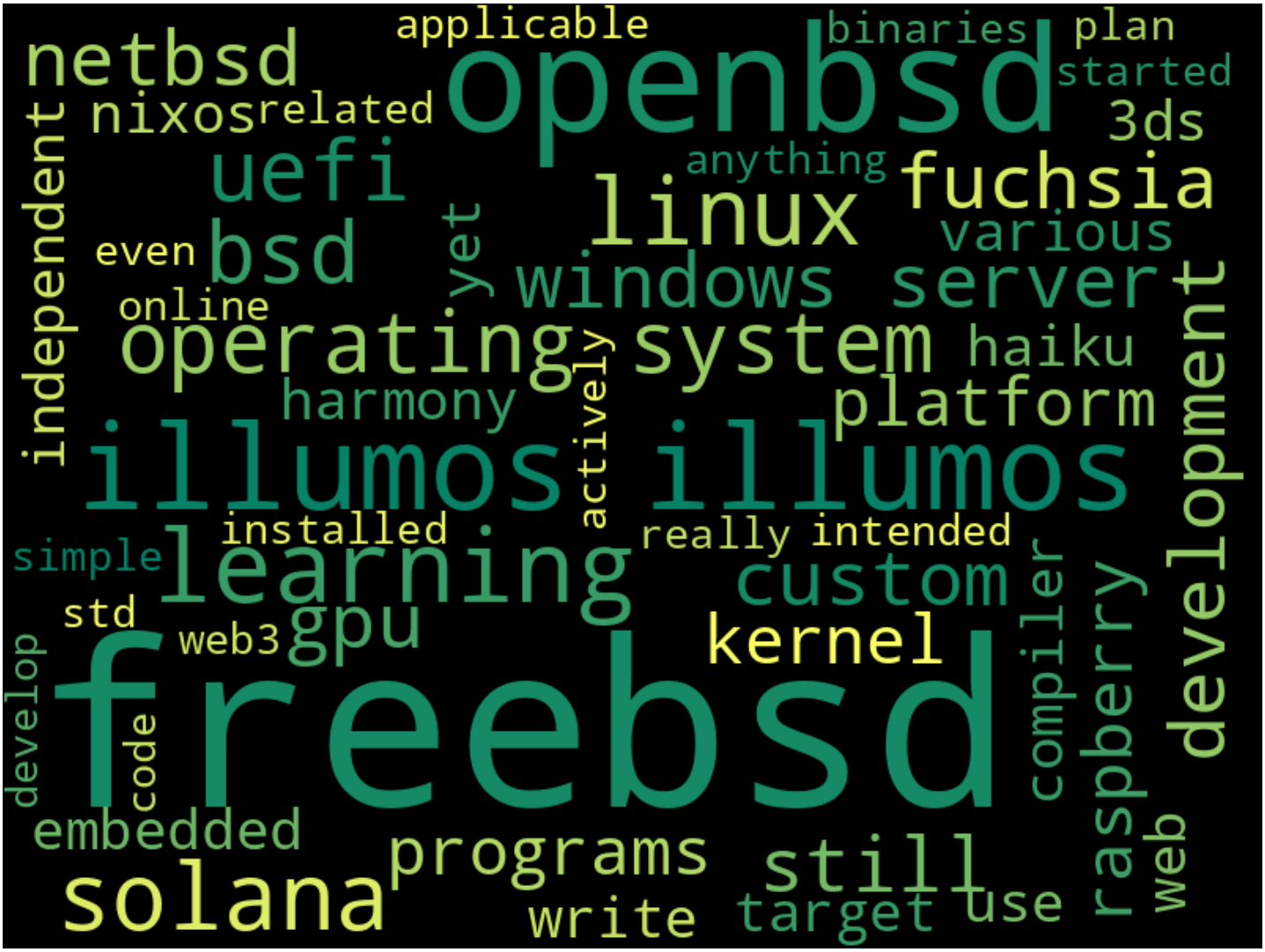

Which operating systems do you use regularly for Rust development?
(total responses = 7976, multiple answers)
Year
2022
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
68.0%
69.7%
73.7%
75.2%
32.8%
33.5%
32.4%
34.1%
30.0%
31.9%
29.8%
27.3%
15.0%
13.8%
13.3%
12.3%
1.1%
1.2%
1.4%
1.5%
Linux
macOS
Windows
Windows Subsystem for Linux
Other

Wordcloud of open answers for the previous chart:



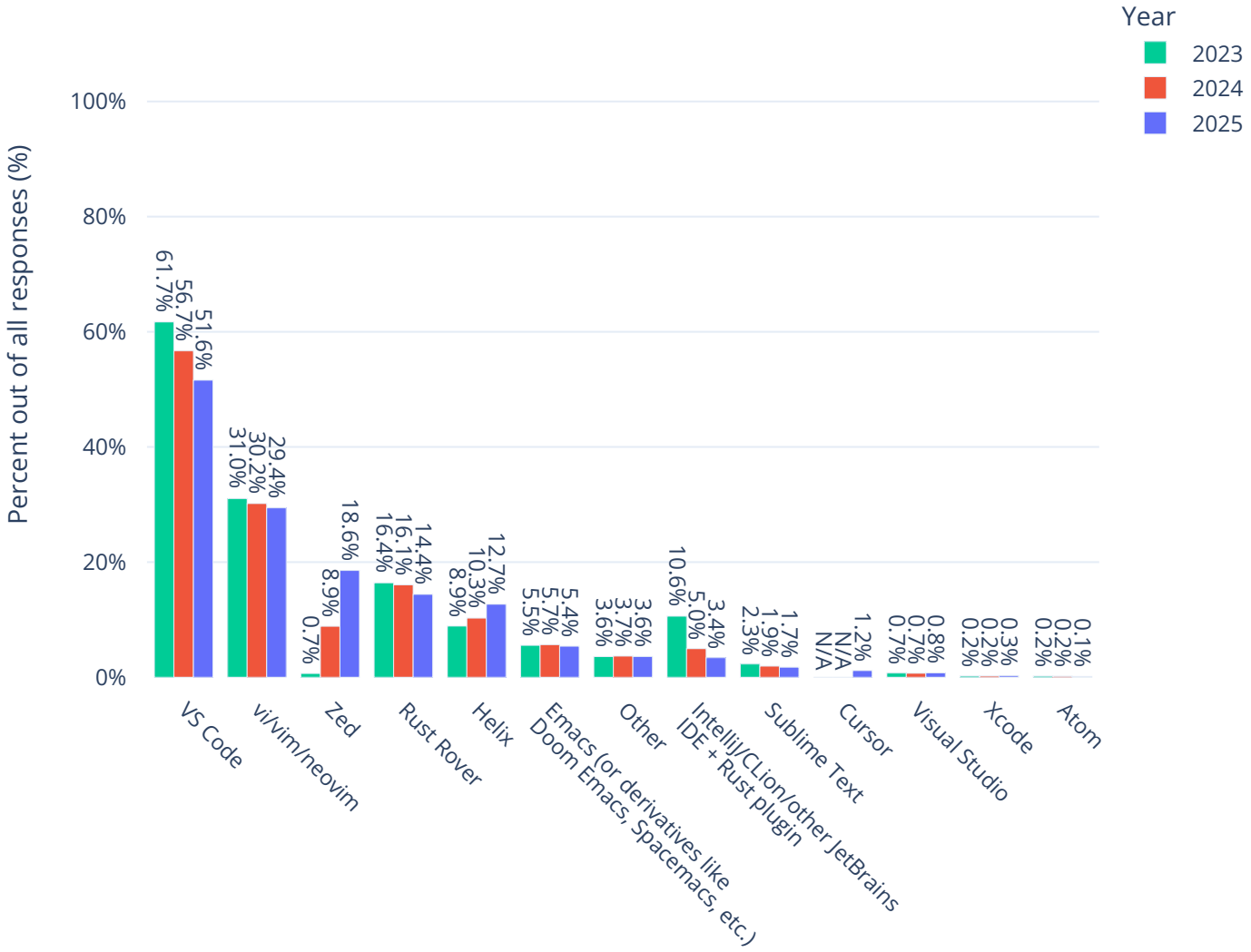

Which operating systems or runtimes do you develop Rust software for?
(total responses = 7926, multiple answers)
Year
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
87.2%
88.4%
44.9%
43.3%
29.7%
30.7%
23.0%
23.9%
16.8%
18.4%
16.7%
17.9%
12.9%
13.2%
7.0%
7.8%
7.7%
7.7%
3.8%
3.9%
1.9%
2.4%
Linux (desktop or server)
Windows
macOS
WebAssembly (for browsers)
Embedded platforms (bare metal)
Explicitly platform-independent (e.g., a library which does not interact with the operating system)
Embedded platforms (with an operating system)
Android
WebAssembly (for other hosts)
iOS
Other

Wordcloud of open answers for the previous chart:

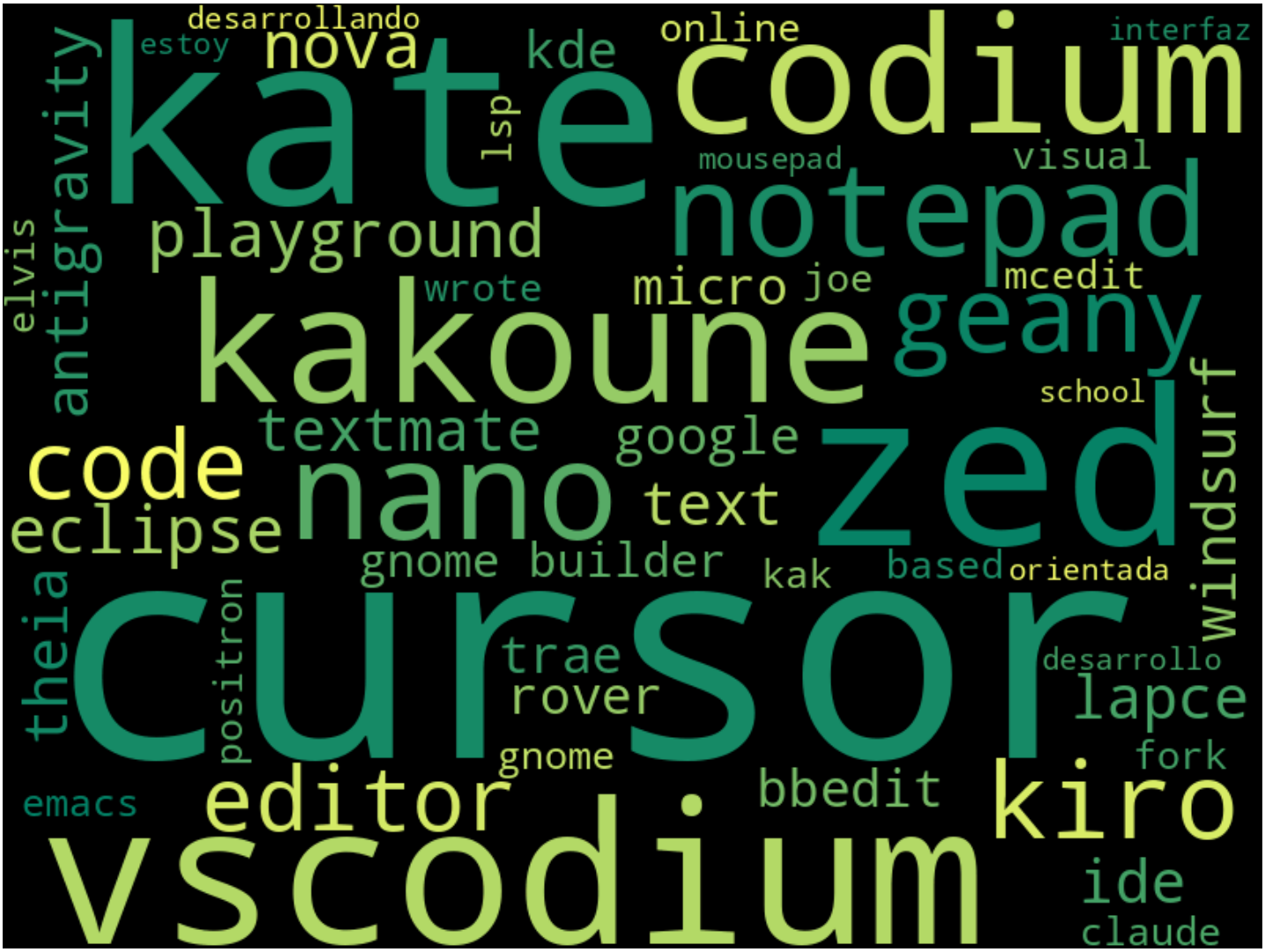


Which editor or IDE setup do you use with Rust code on a regular basis?

(total responses = 7967, multiple answers)

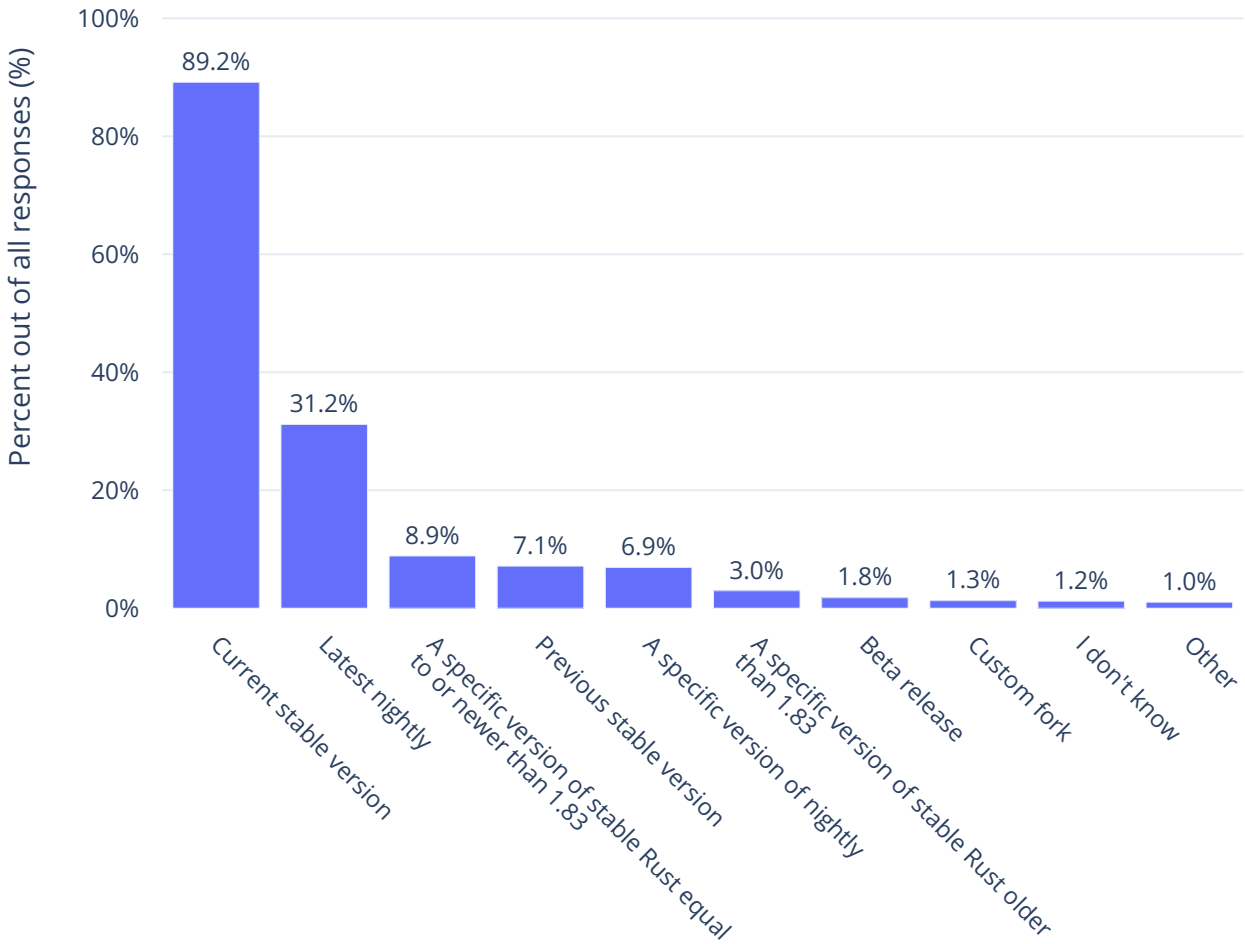


Wordcloud of open answers for the previous chart:

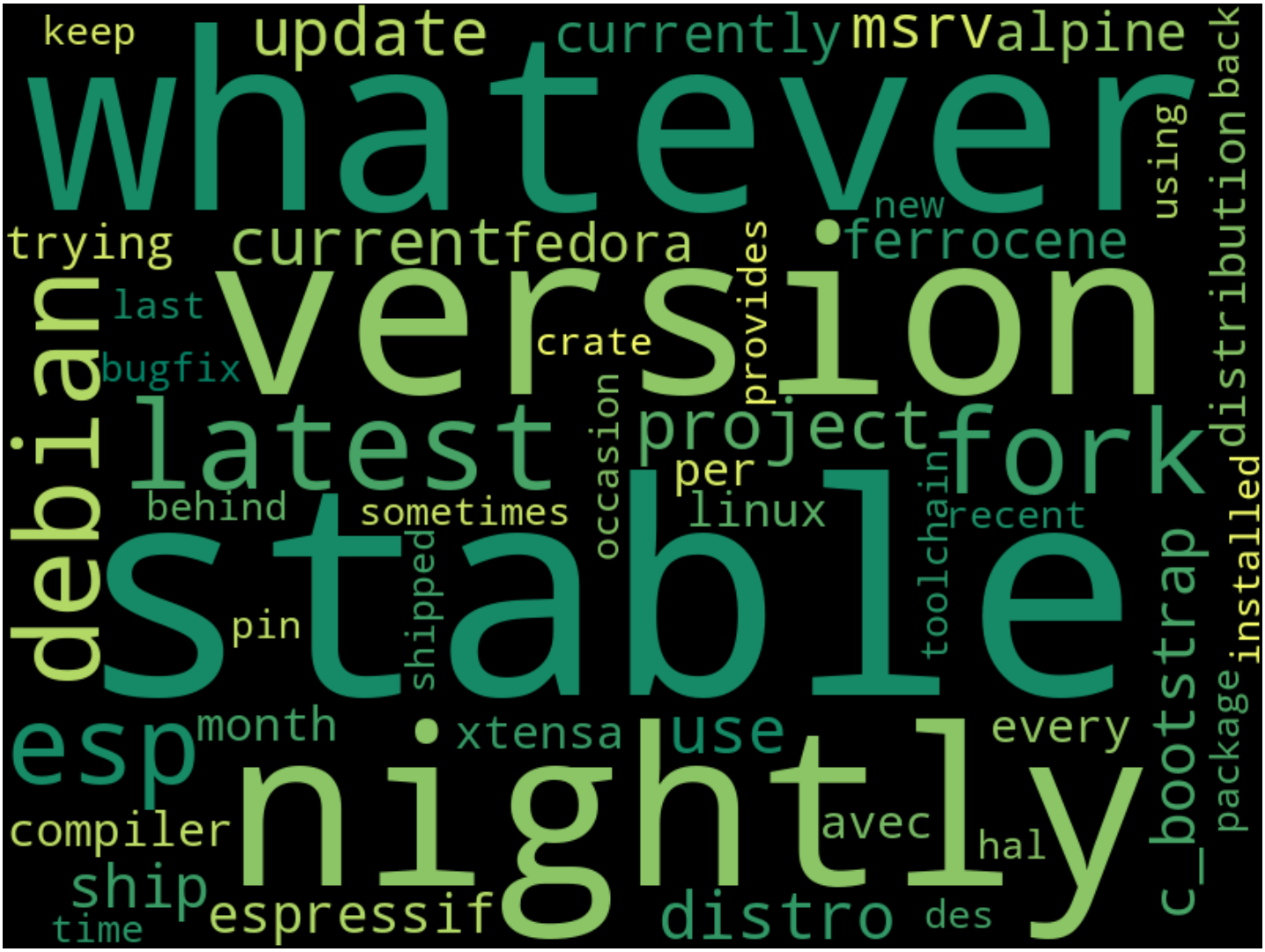


Which version(s) of Rust do you use for development?

(total responses = 7969, multiple answers)

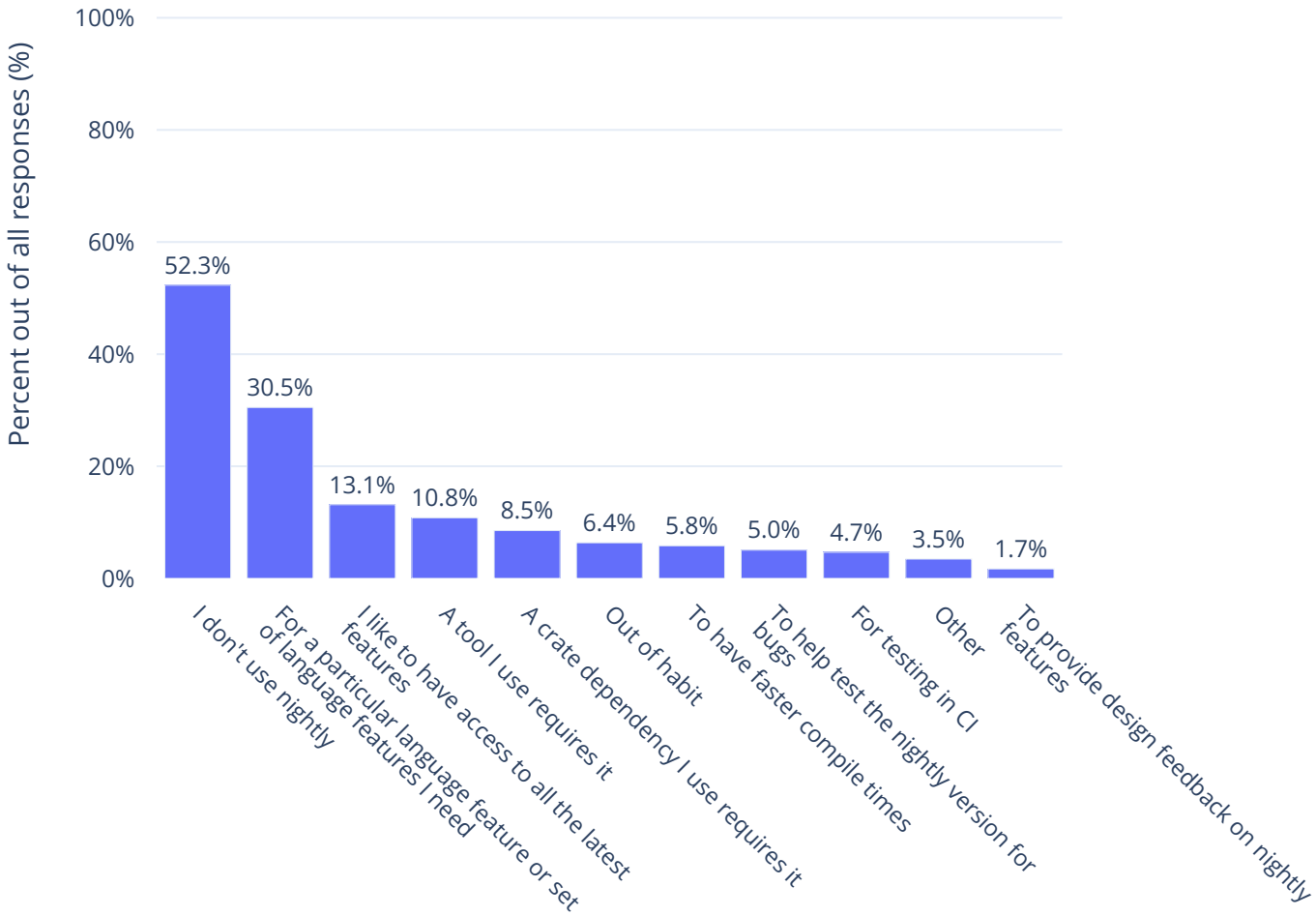


Wordcloud of open answers for the previous chart:

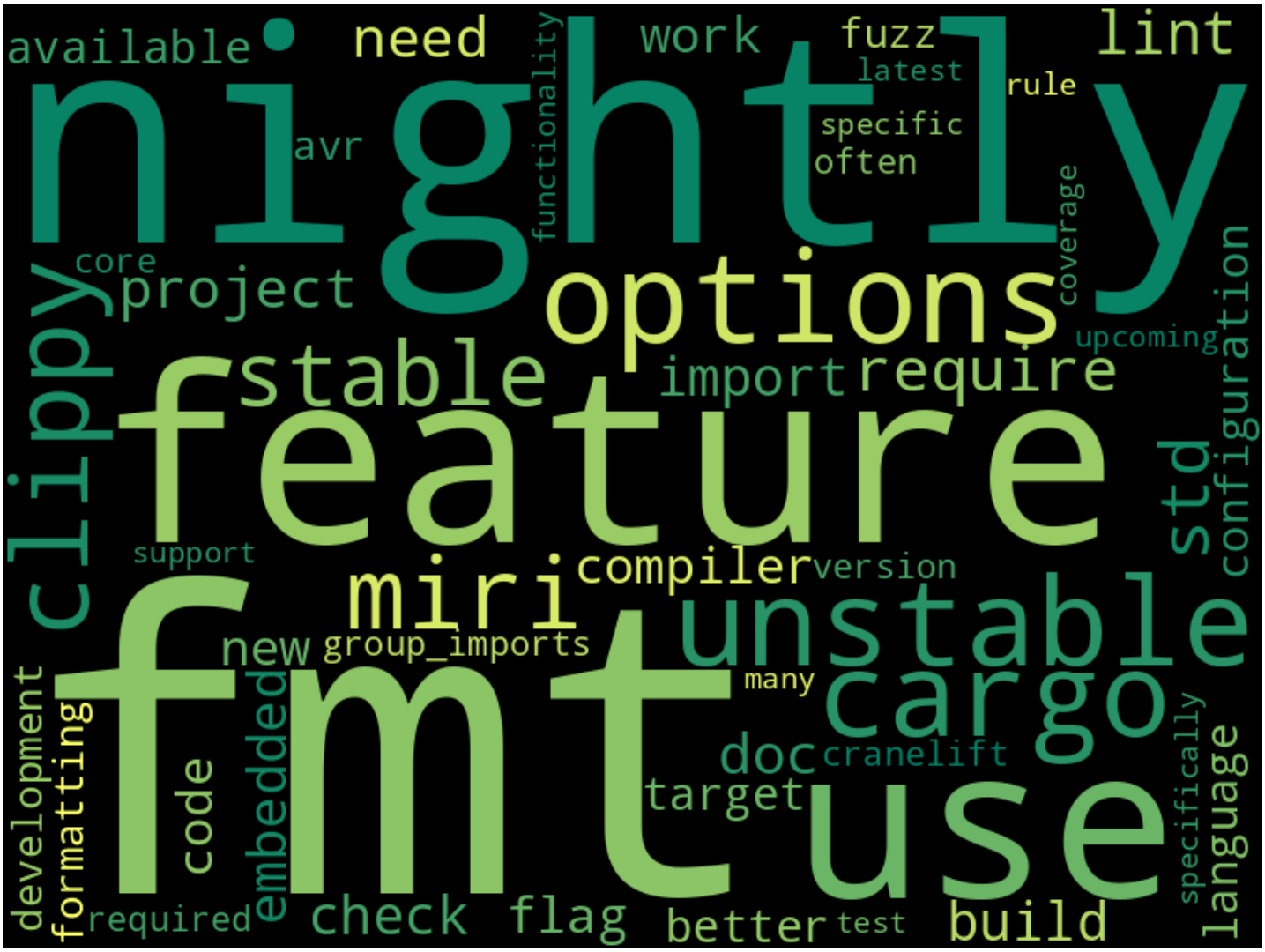


If you use nightly, why?

(total responses = 7272, multiple answers)

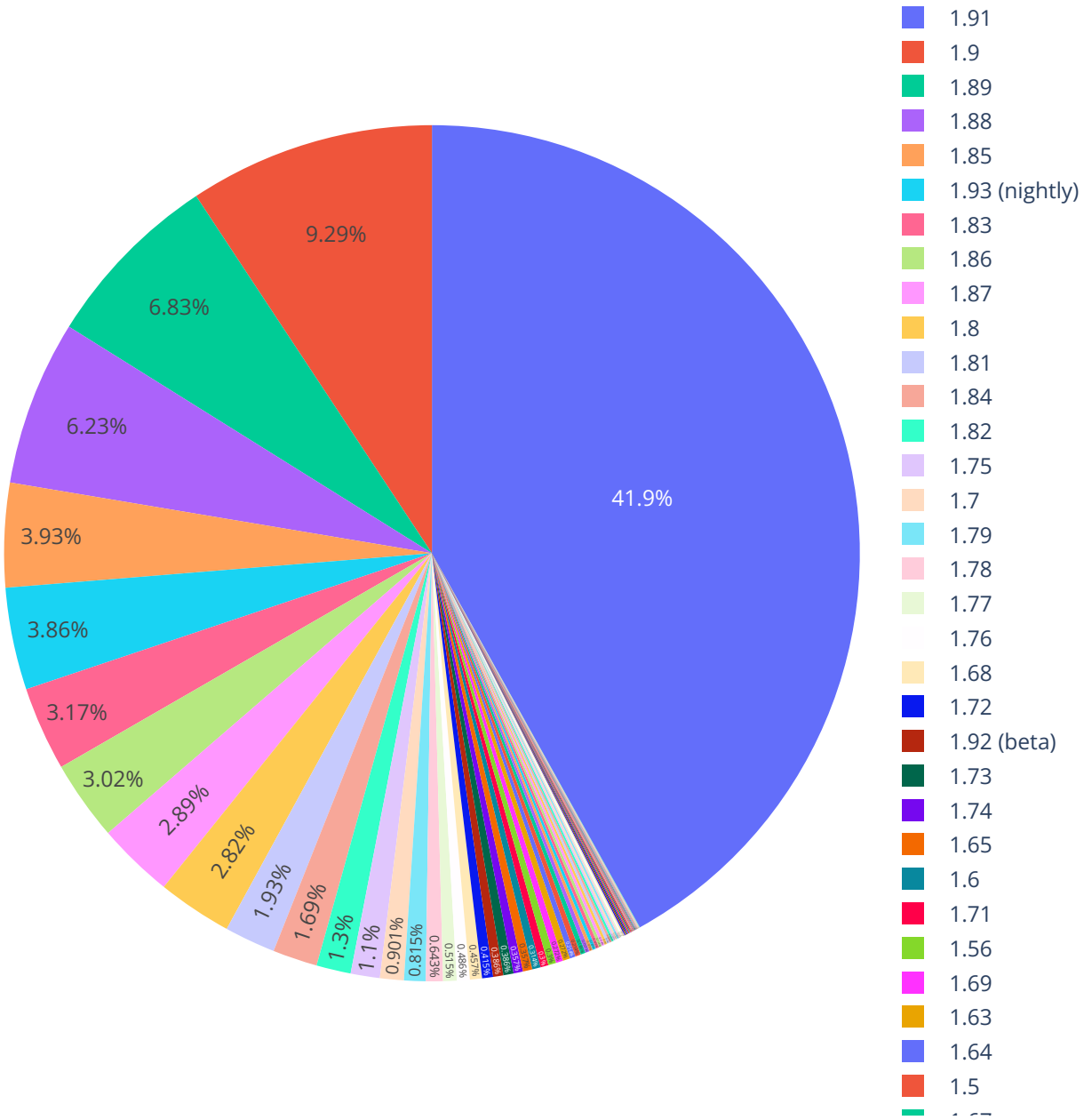


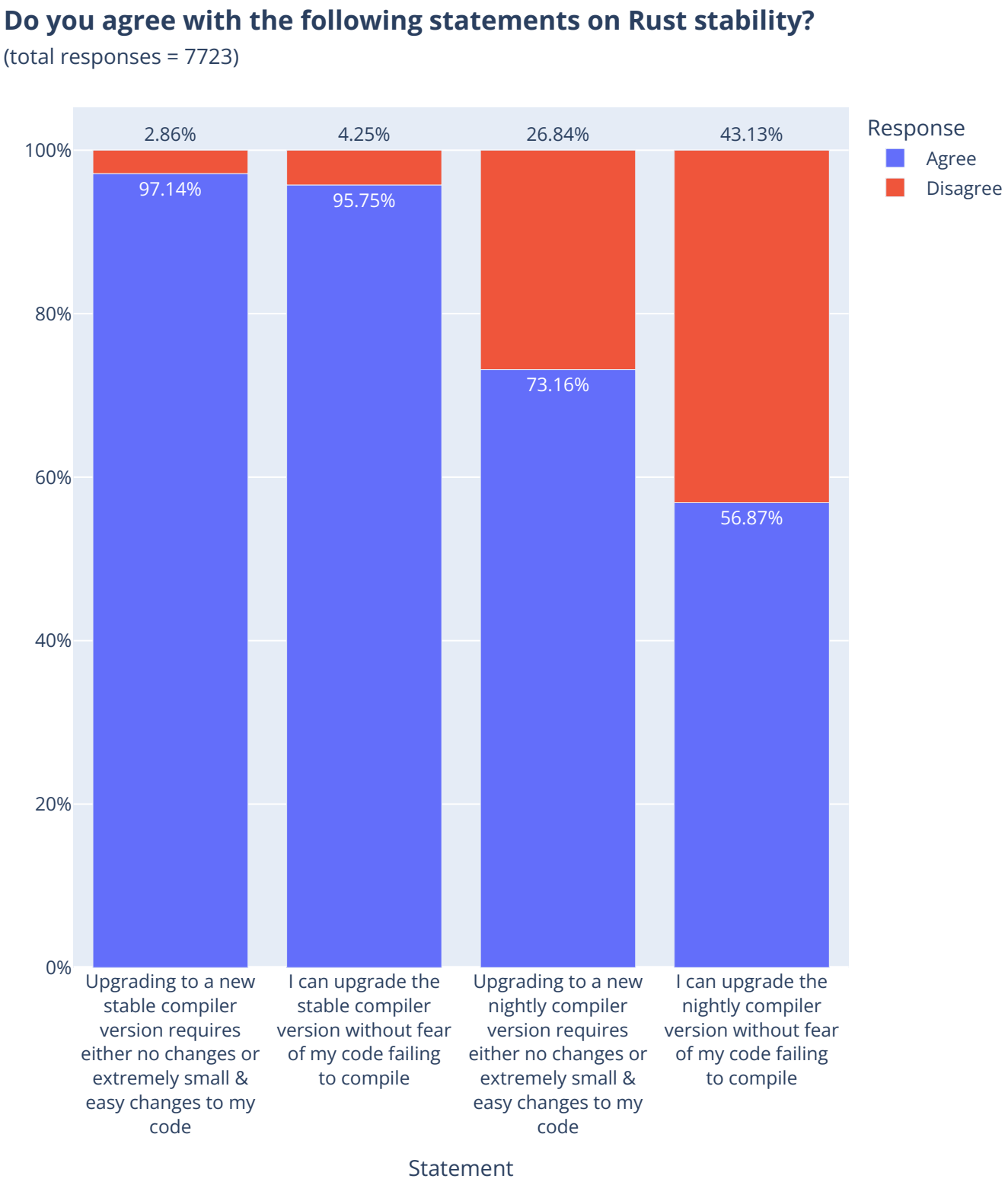
Wordcloud of open answers for the previous chart:



What is the oldest version of Rust you use for any development task?

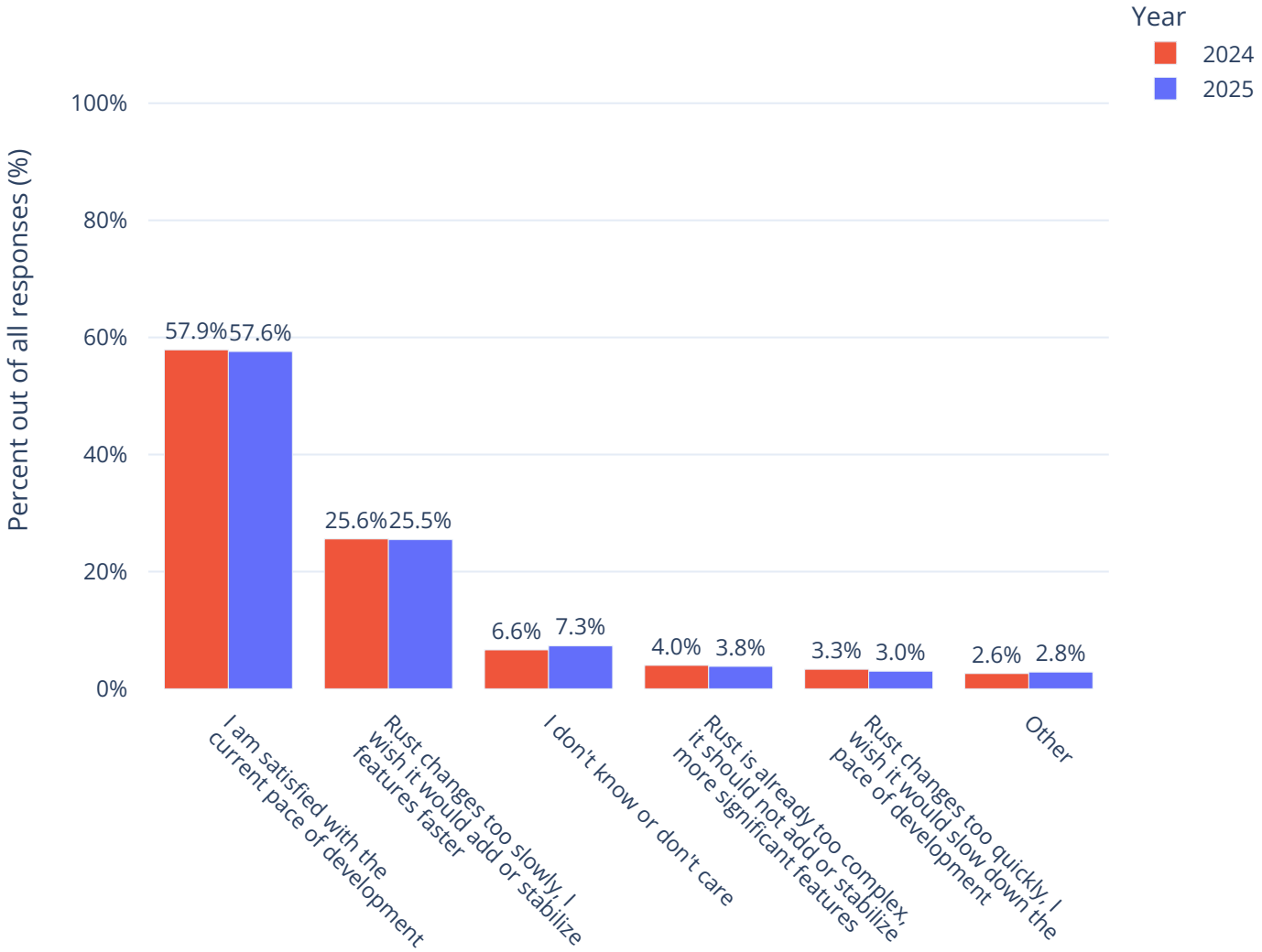
(total responses = 6996)




Do you agree with the following statements on Rust stability?
(total responses = 7723)
2.86%
4.25%
26.84%
43.13%
97.14%
95.75%
73.16%
56.87%
100%
80%
60%
40%
20%
0%
Upgrading to a new stable compiler version requires either no changes or extremely small & easy changes to my code
I can upgrade the stable compiler version without fear of my code failing to compile
Upgrading to a new nightly compiler version requires either no changes or extremely small & easy changes to my code
I can upgrade the nightly compiler version without fear of my code failing to compile
Statement
Response
Agree
Disagree

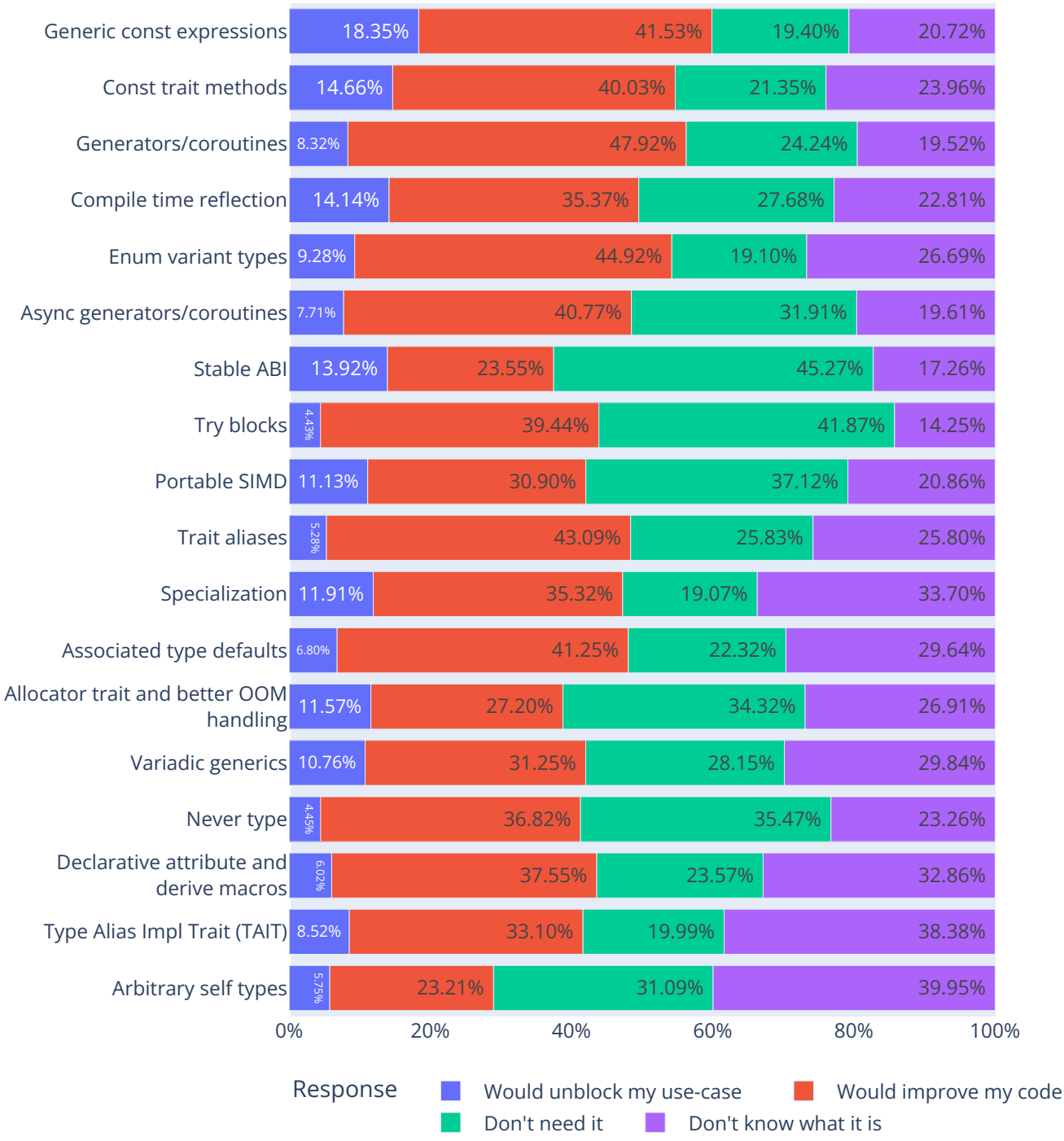
What is your opinion on how fast Rust evolves?

(total responses = 6808, single answer)

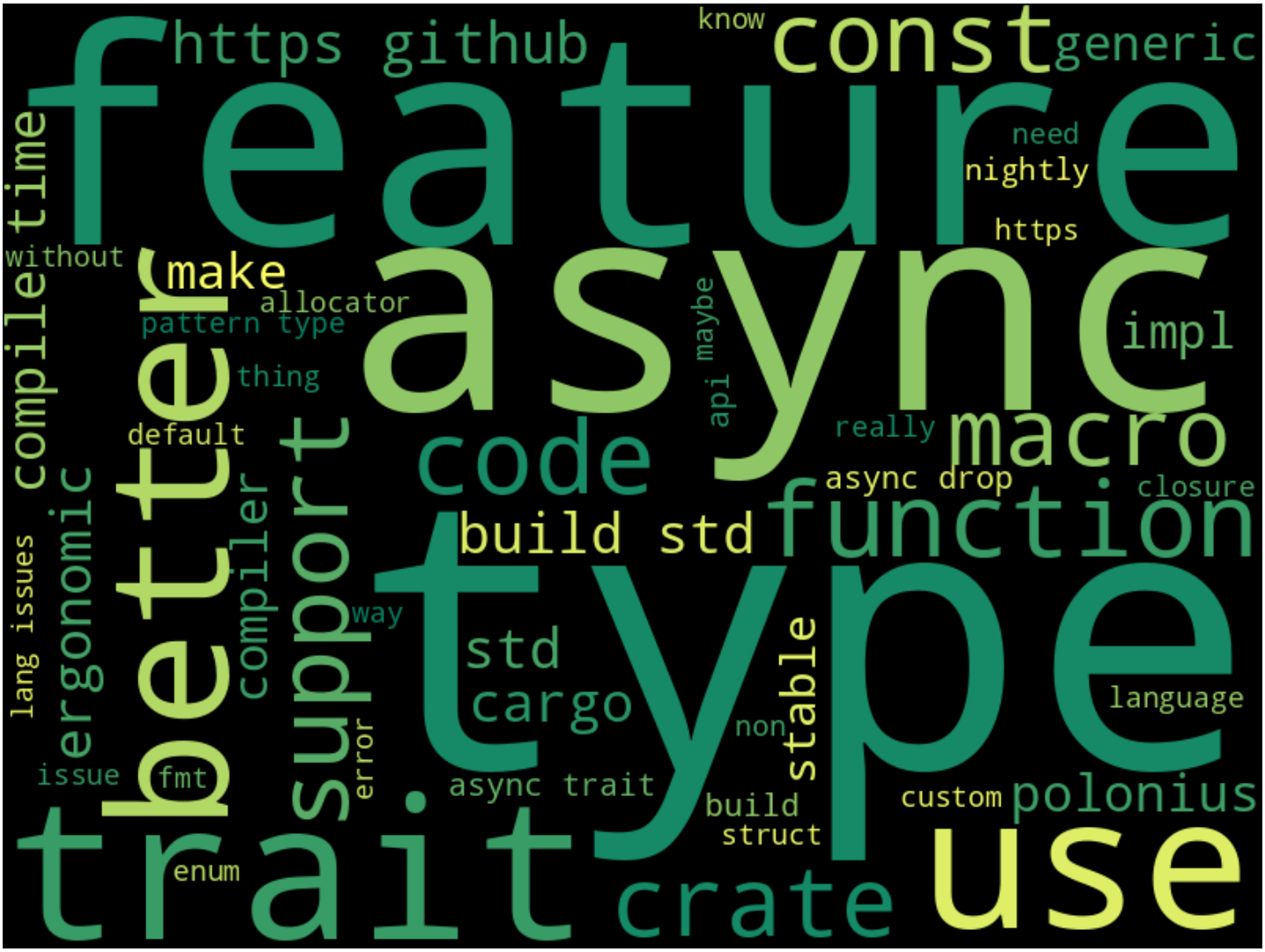


Which unimplemented (or nightly only) features are you looking for to be stabilized?

(total responses = 6514)

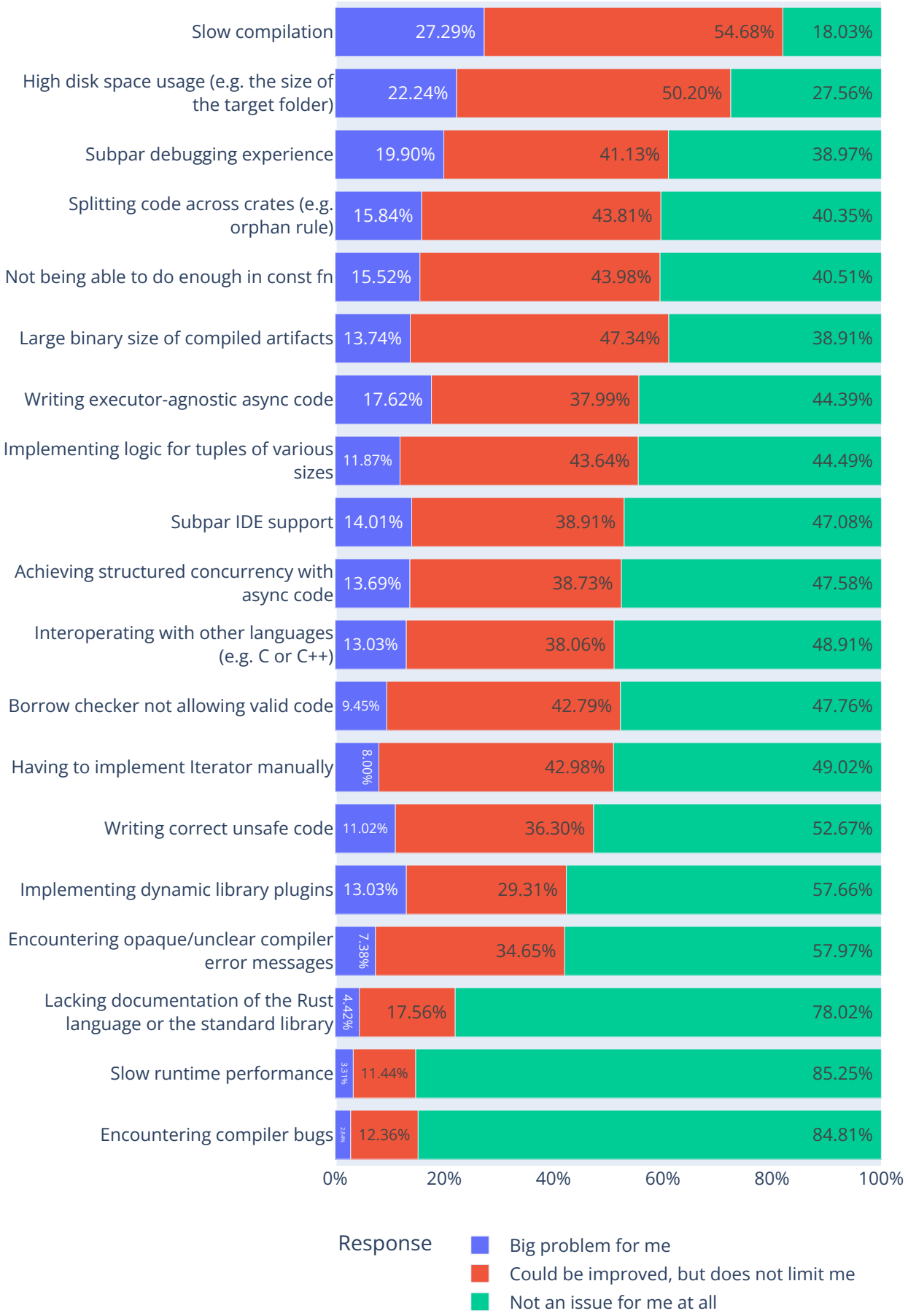


Wordcloud of open answers for the previous chart:

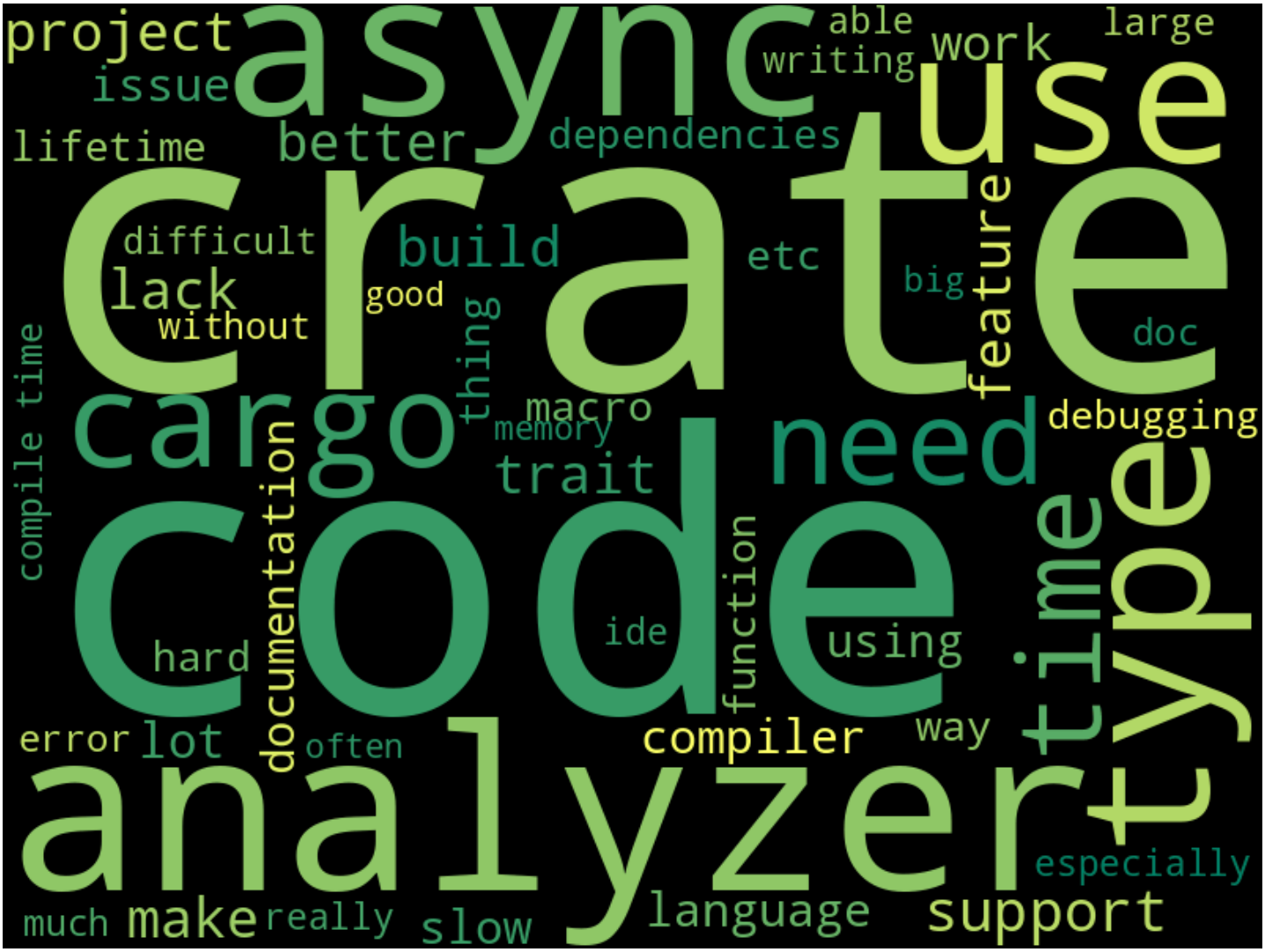


Which of the following aspects of Rust present non-trivial problems to your programming productivity?

(total responses = 6653)

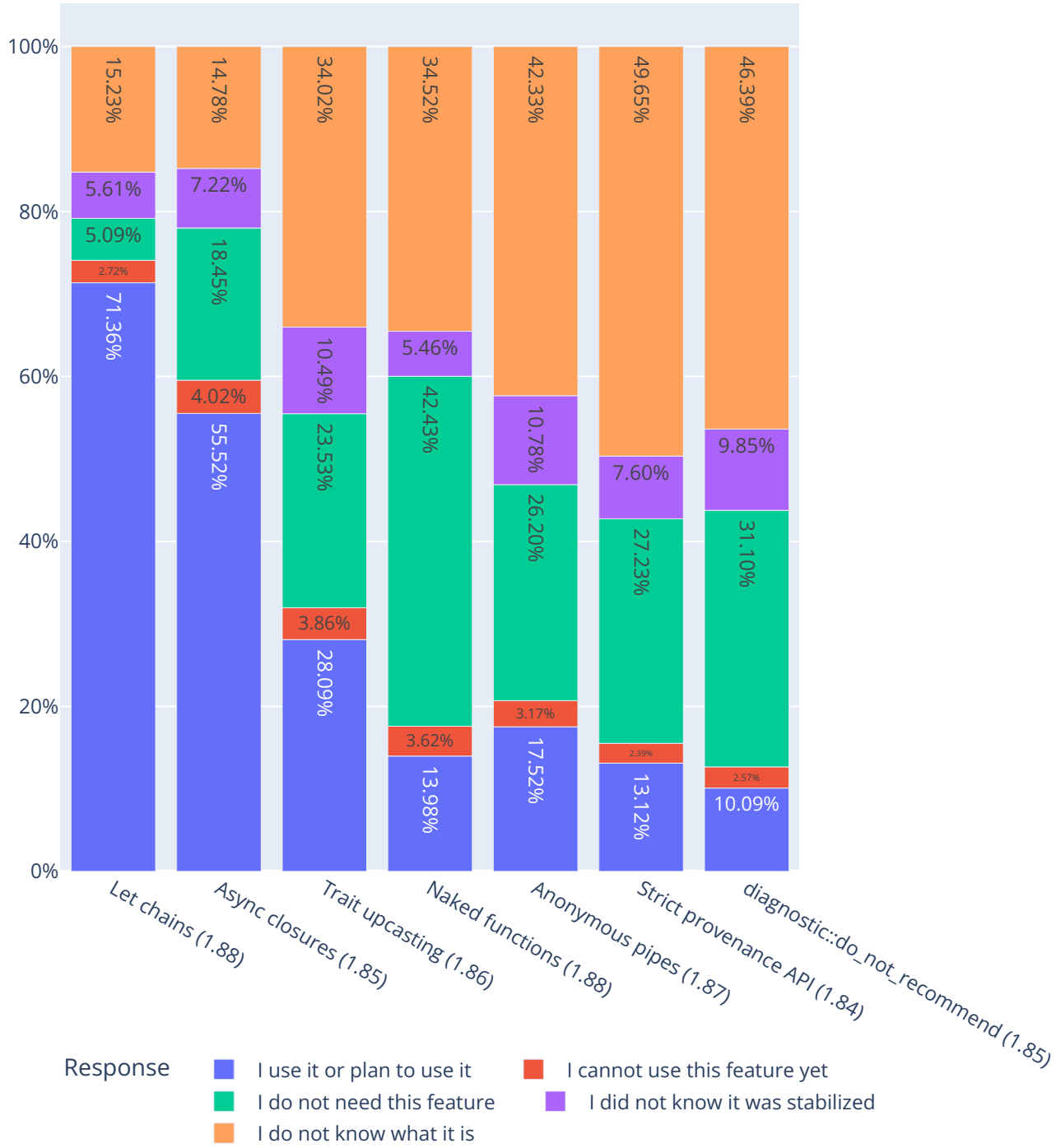


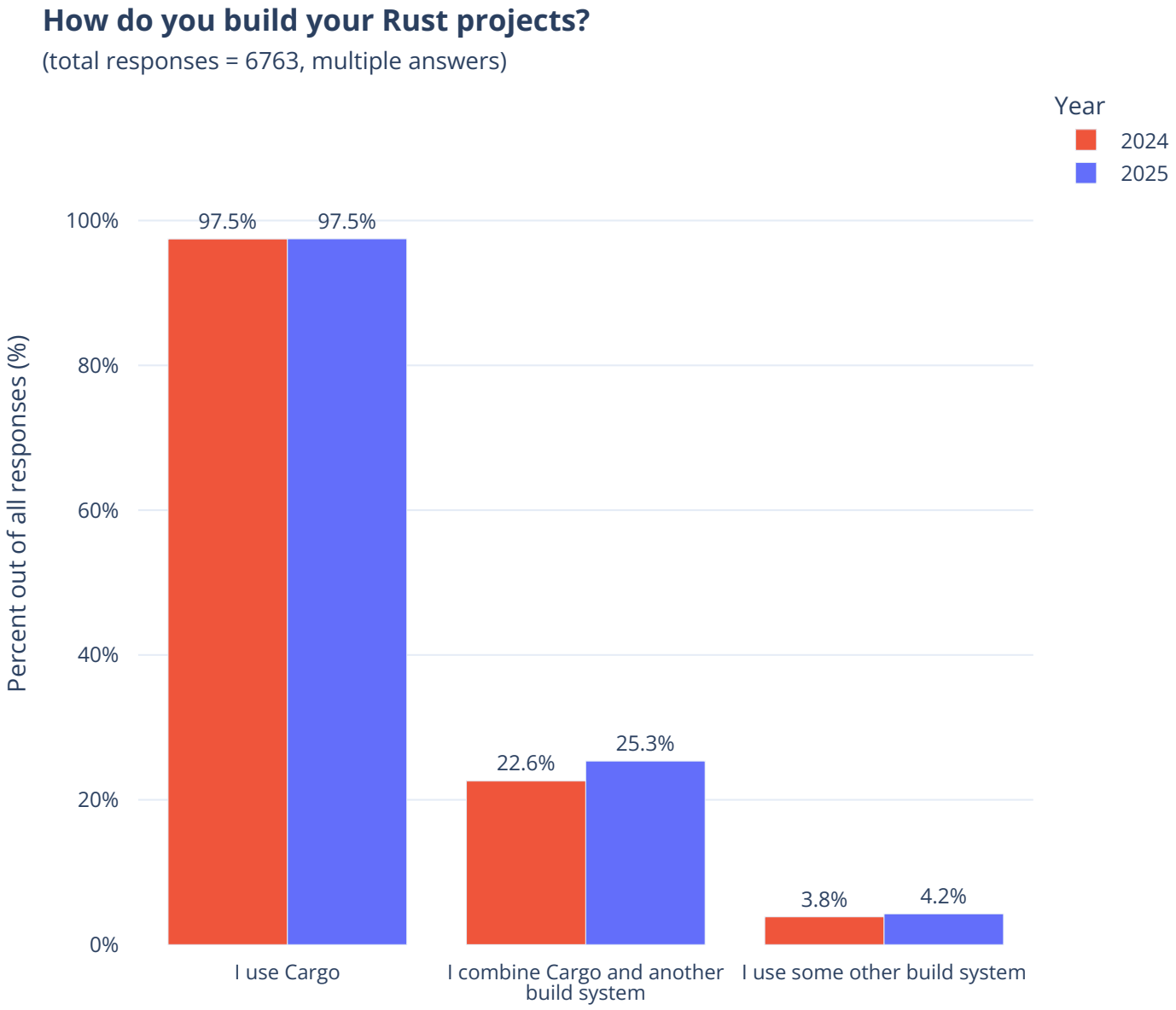
Wordcloud of open answers for the previous chart:



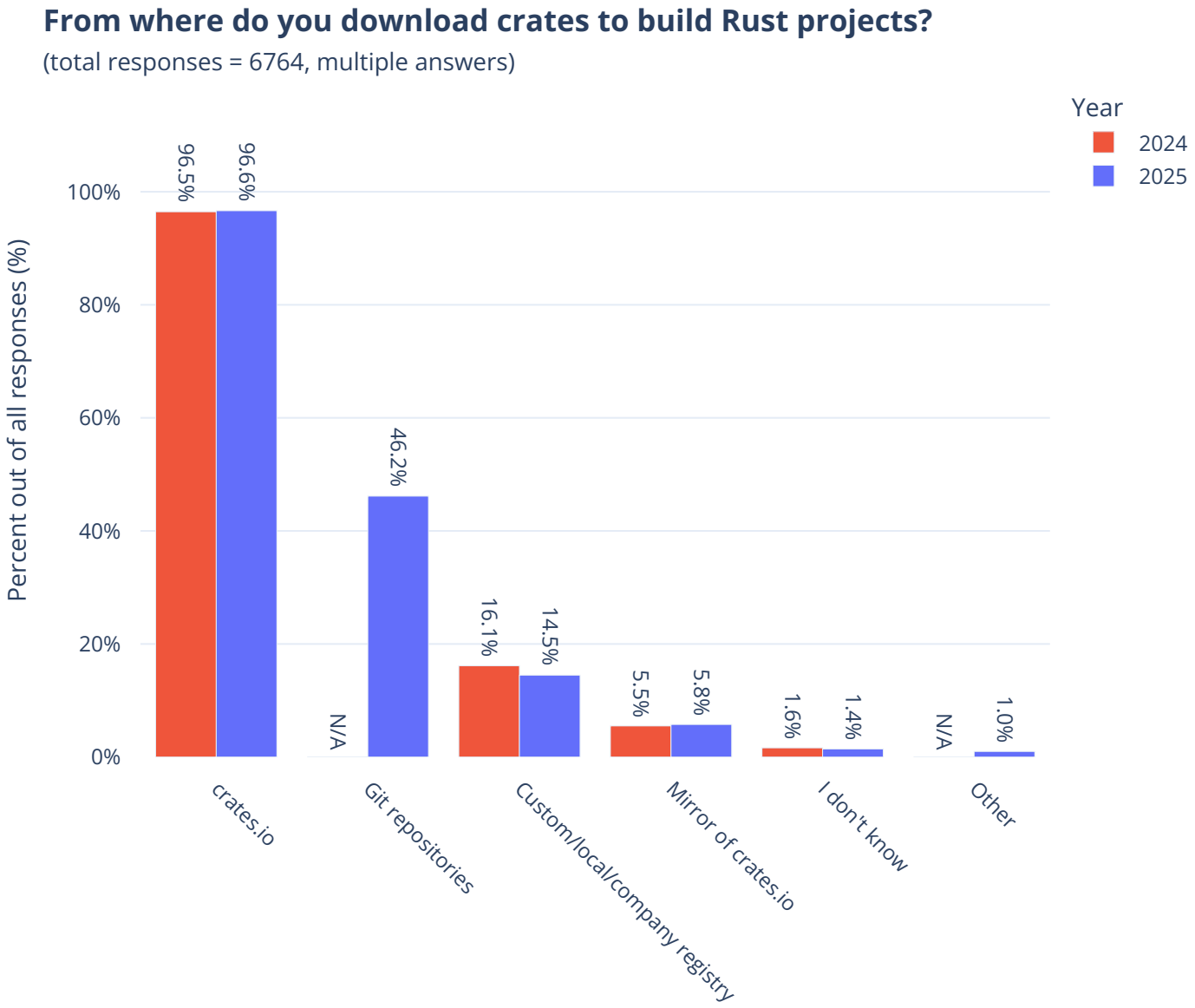
Which features stabilized in the past 12 months do you use the most?

(total responses = 6400)




How do you build your Rust projects?
(total responses = 6763, multiple answers)
Year
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
97.5%
97.5%
22.6%
25.3%
3.8%
4.2%
I use Cargo
I combine Cargo and another build system
I use some other build system

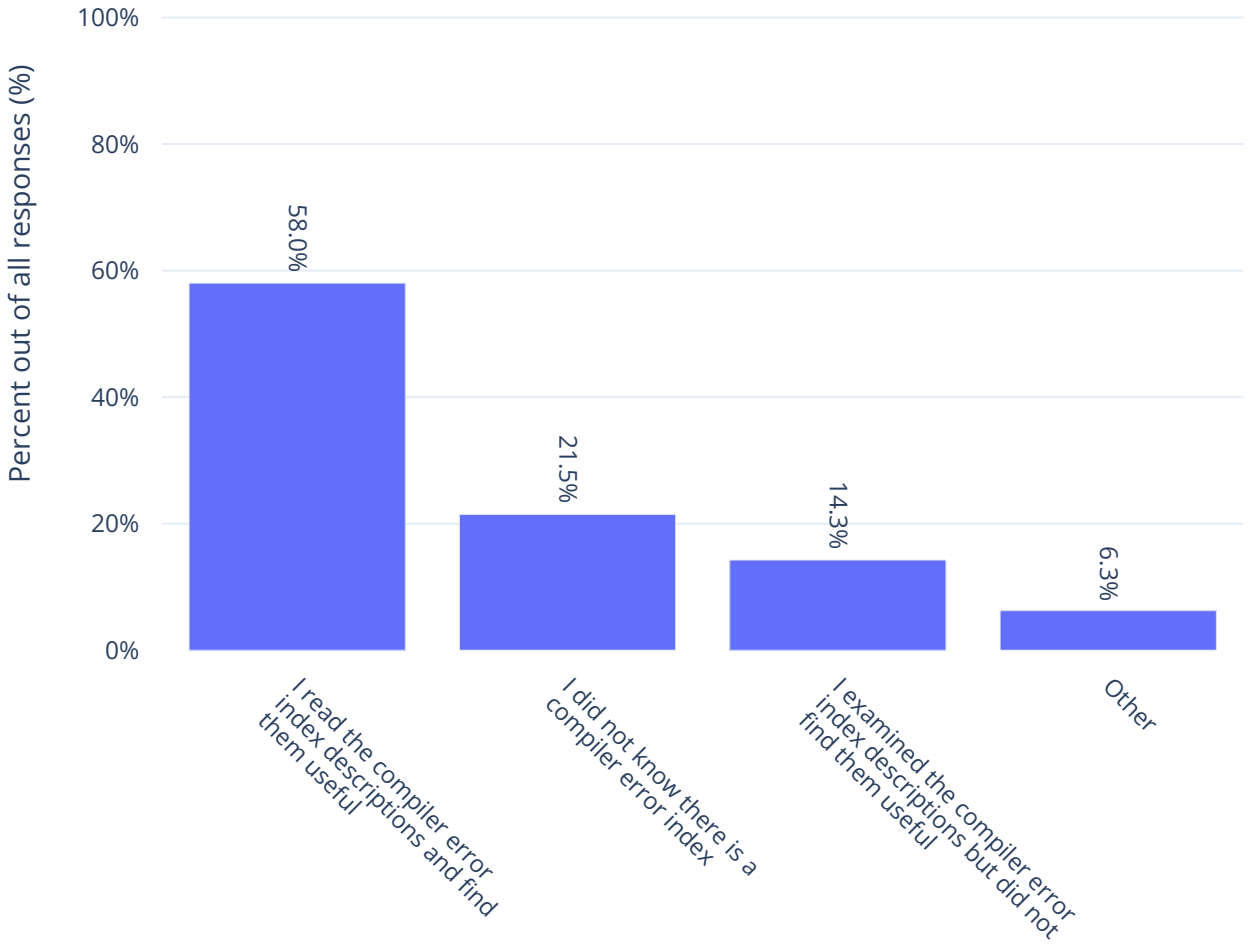
Wordcloud of open answers for the previous chart:


From where do you download crates to build Rust projects?
(total responses = 6764, multiple answers)
Year
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
96.5%
96.6%
N/A
46.2%
16.1%
14.5%
5.5%
5.8%
1.6%
1.4%
N/A
1.0%
crates.io
Git repositories
Custom/local/company registry
Mirror of crates.io
I don't know
Other

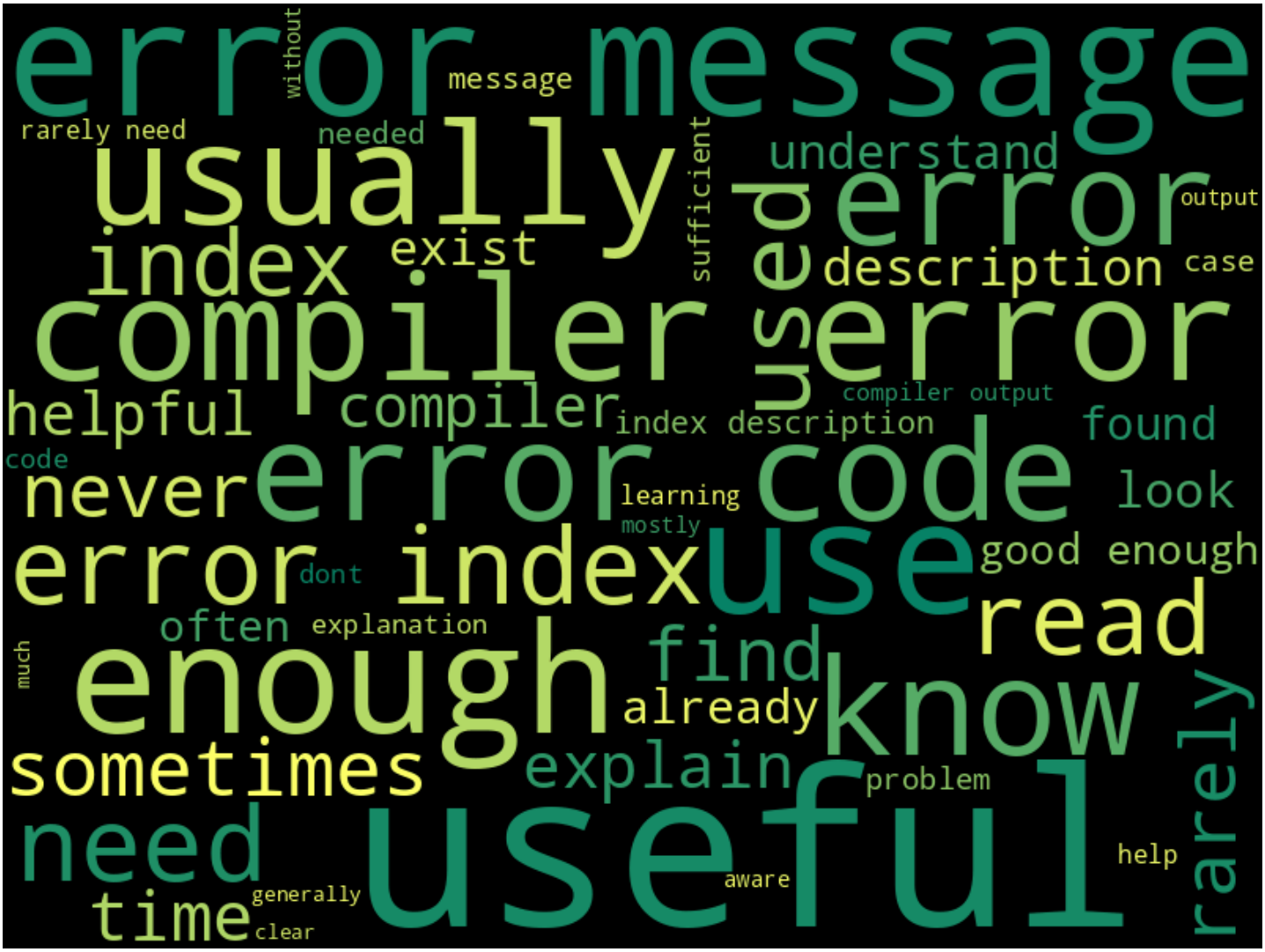
Wordcloud of open answers for the previous chart:

Do you make use of compiler error codes?

(total responses = 6606, single answer)

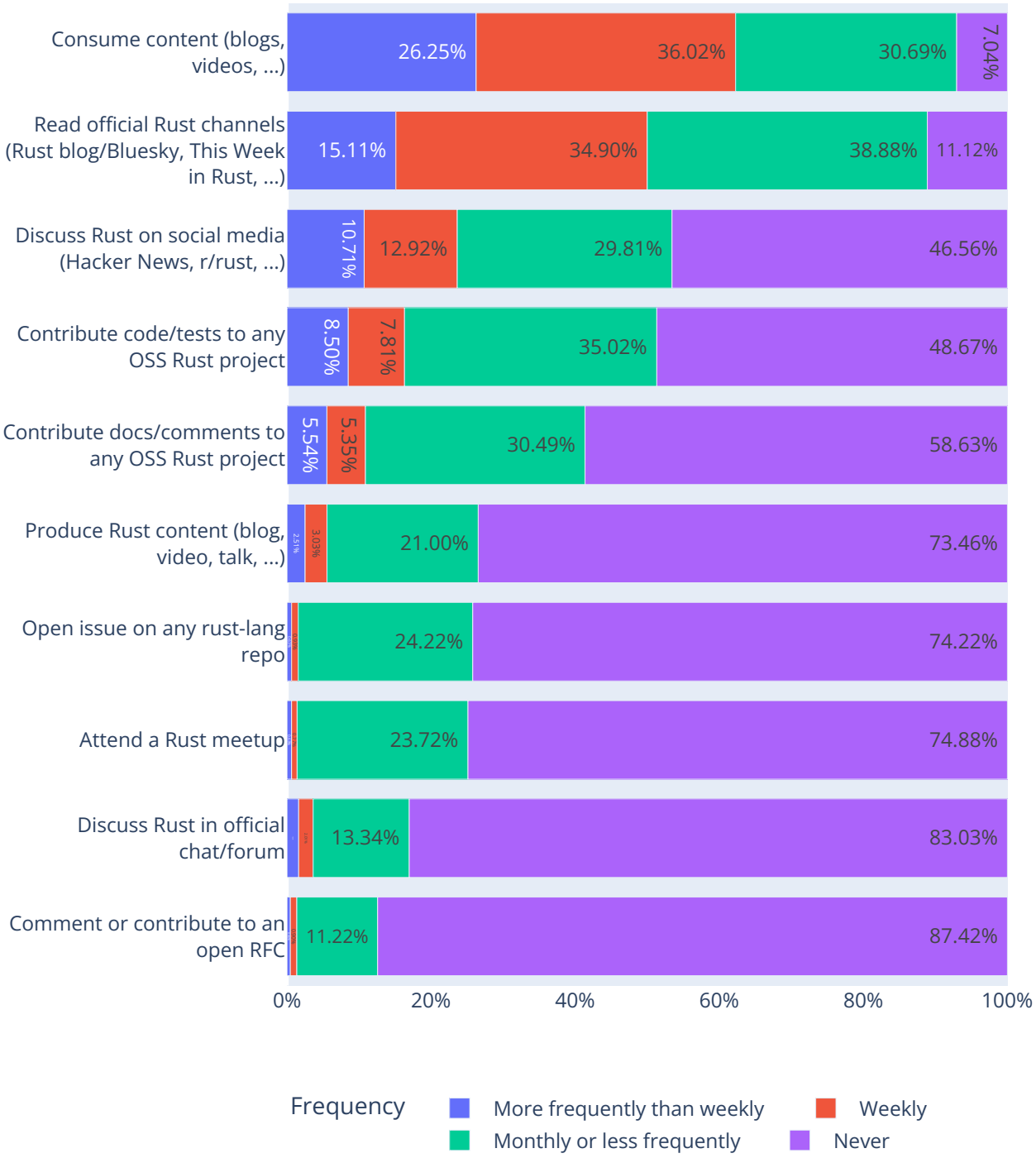


Wordcloud of open answers for the previous chart:



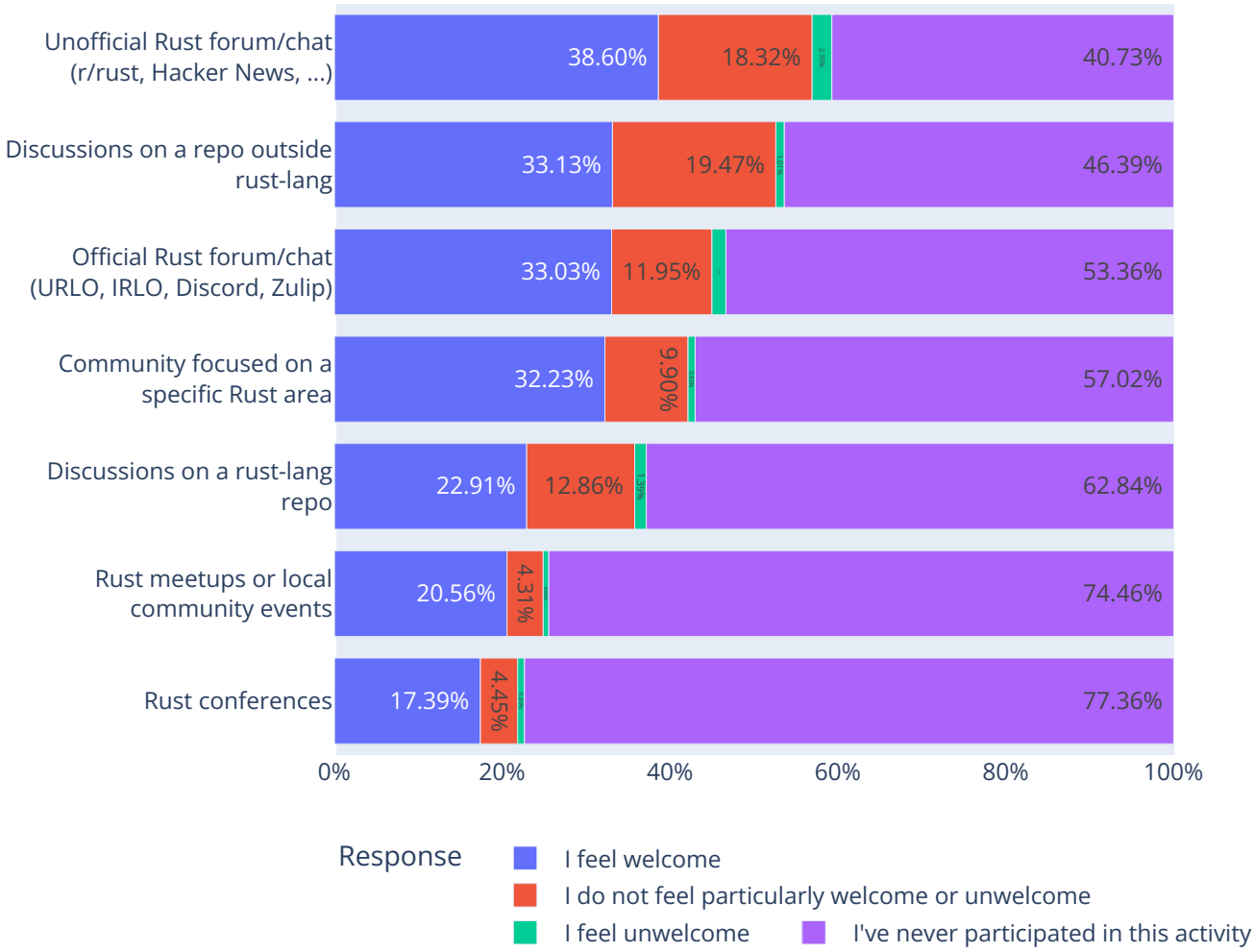
Roughly how often do you engage in the following Rust community activities?

(total responses = 6548)



What has your experience been like in the following Rust community spaces?

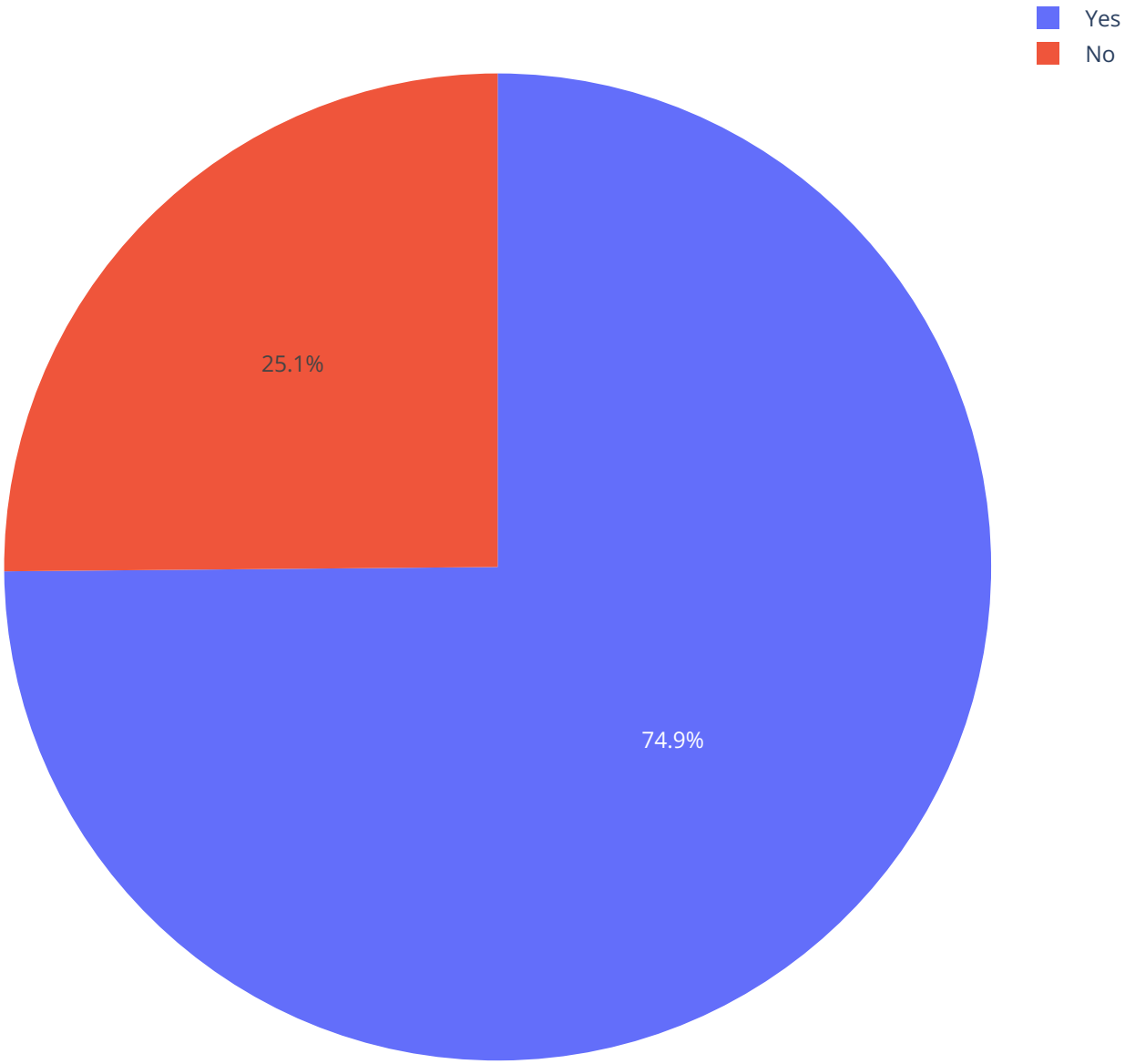
(total responses = 6420)



are-you-employed

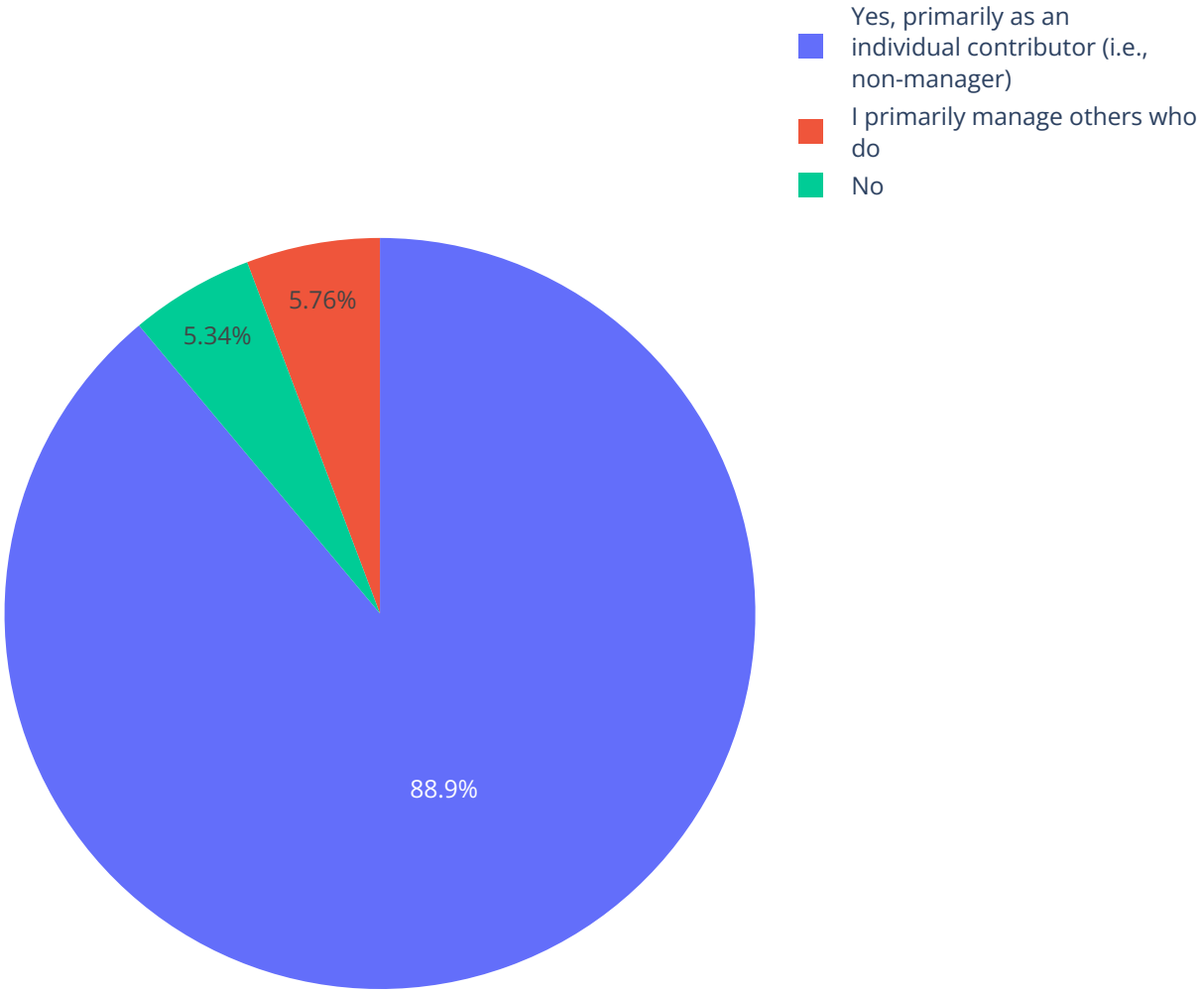
Are you employed full- or part-time (including paid internships)?

(total responses = 7319)



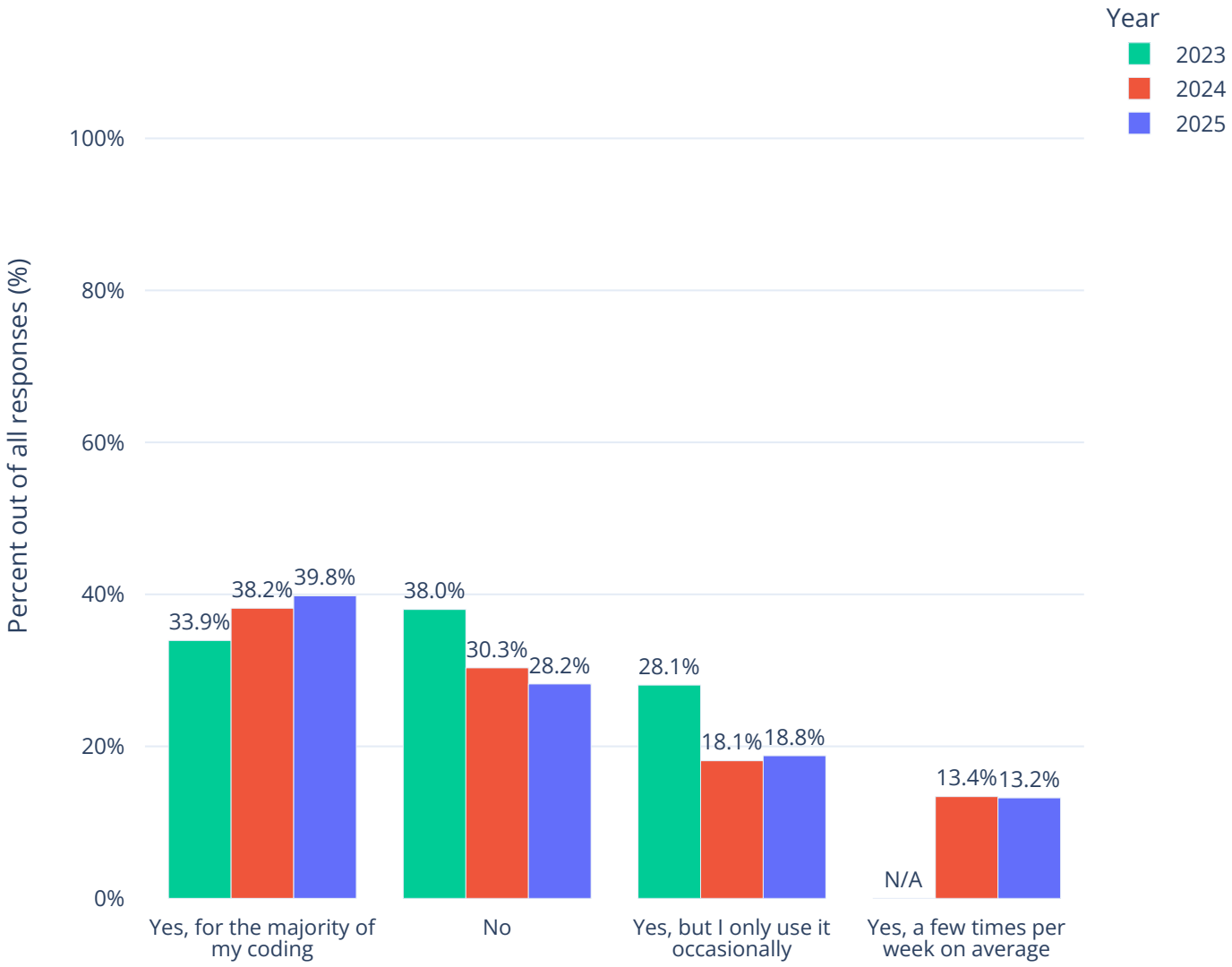
Do you write or design software in your work?

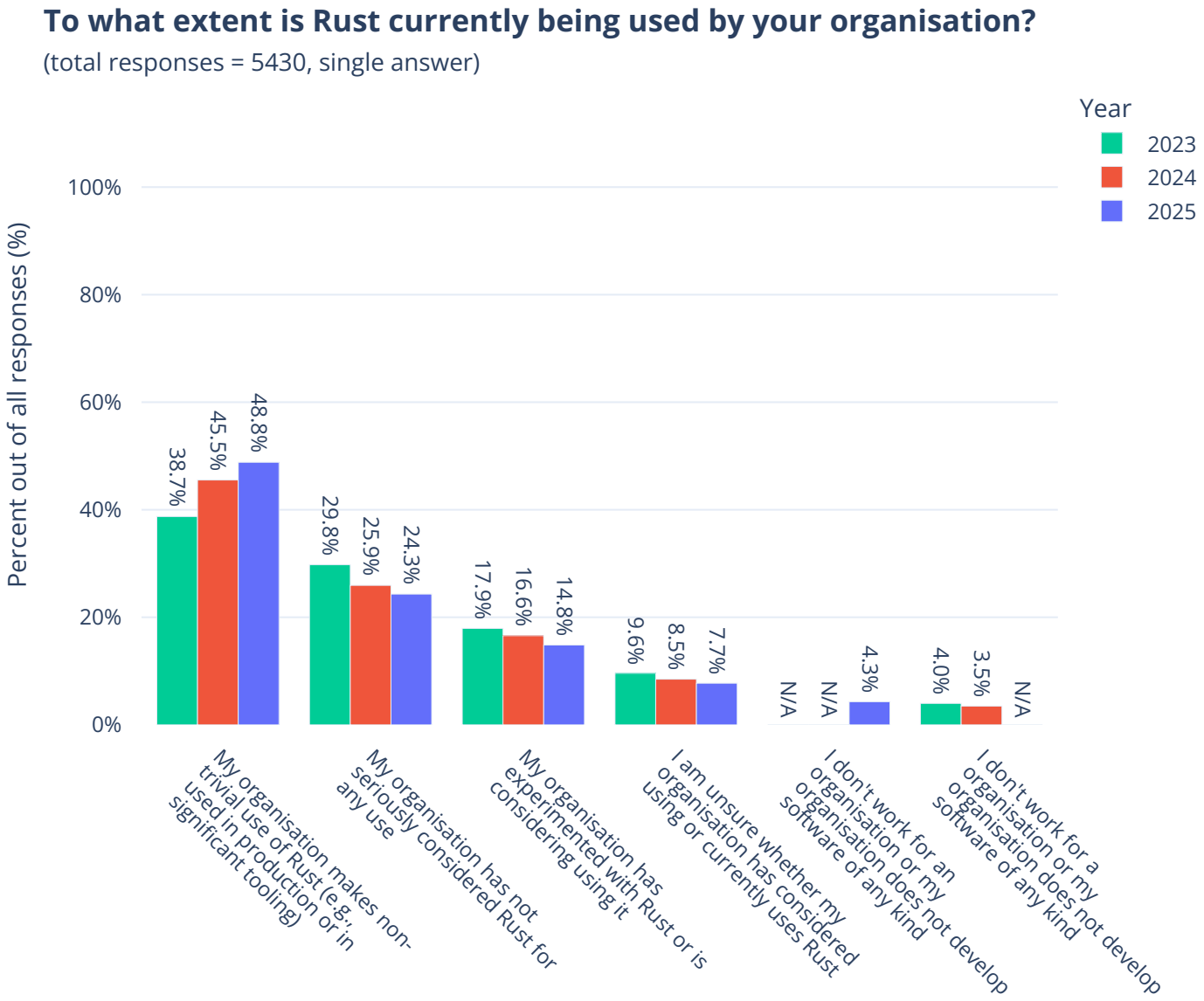
(total responses = 5473)



Are you personally using Rust at work?

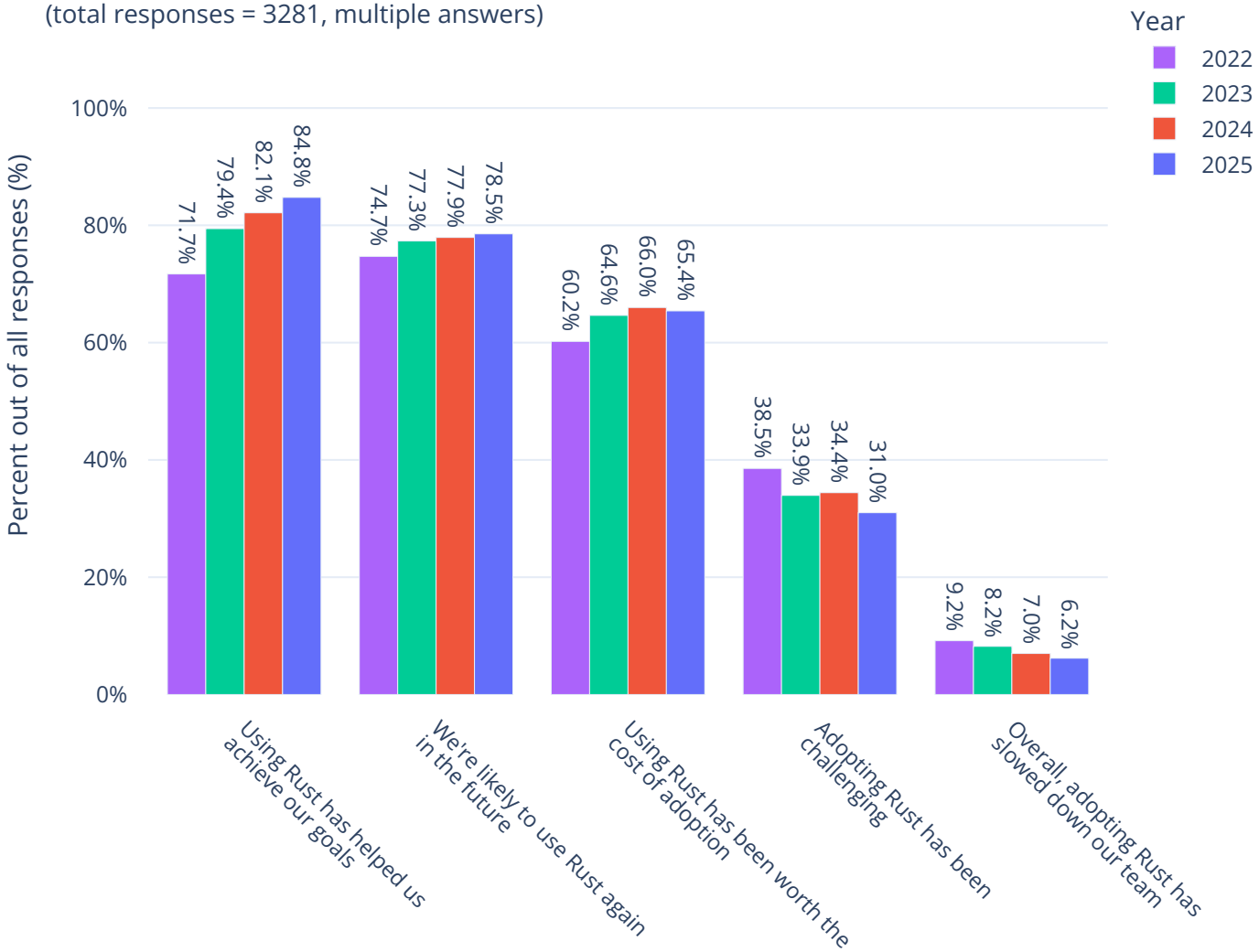
(total responses = 4977, single answer)

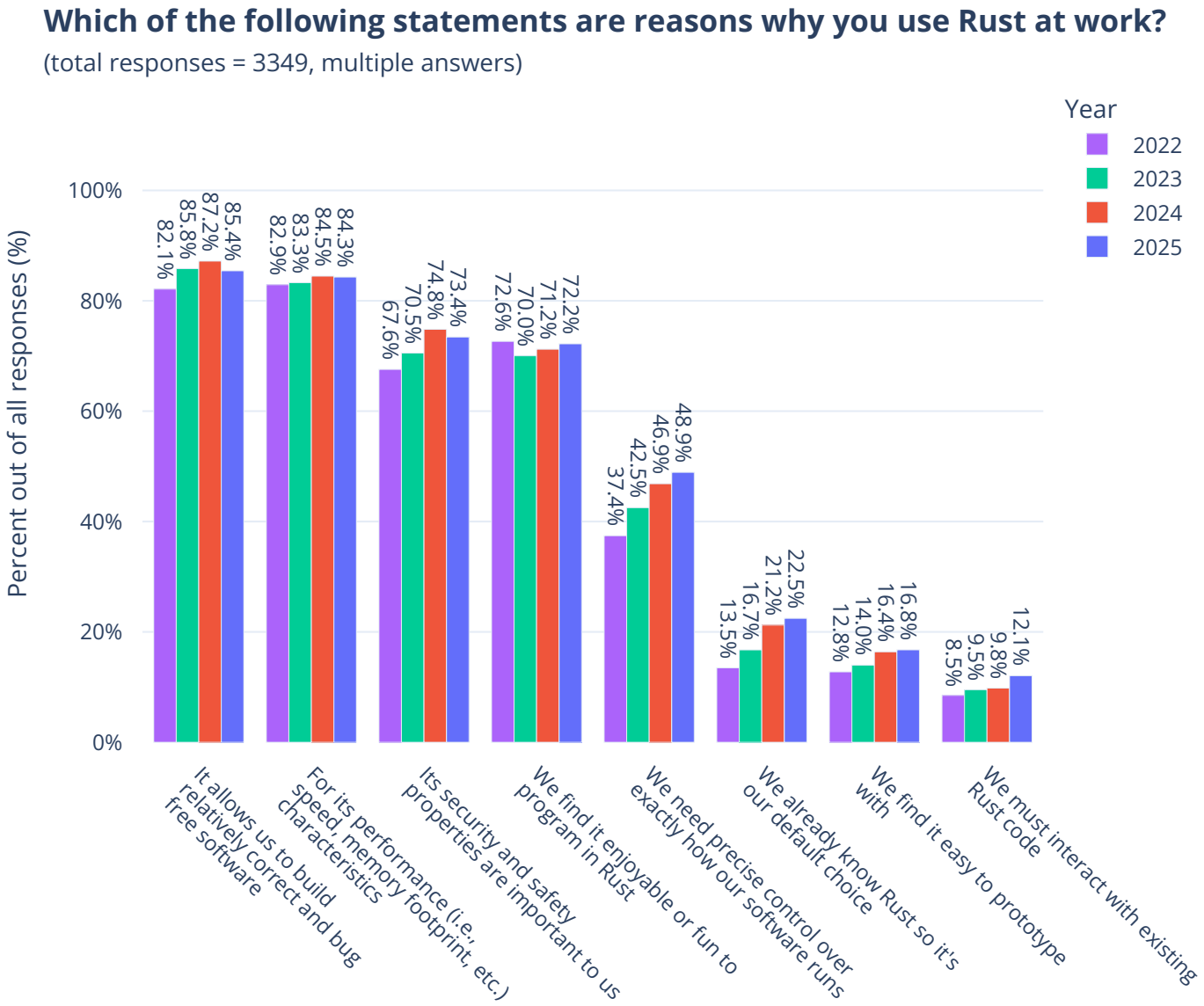



To what extent is Rust currently being used by your organisation?
(total responses = 5430, single answer)
Year
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
38.7%
45.5%
48.8%
29.8%
25.9%
24.3%
17.9%
16.6%
14.8%
9.6%
8.5%
7.7%
N/A
N/A
4.3%
4.0%
3.5%
N/A
My organisation makes non-trivial use of Rust (e.g., used in production or in significant tooling)
My organisation has not seriously considered Rust for any use
My organisation has experimented with Rust or is considering using it
I am unsure whether my organisation has considered using or currently uses Rust
I don't work for an organisation or my organisation does not develop software of any kind
I don't work for a organisation or my organisation does not develop software of any kind

Which of the following statements apply to your experience using Rust at work?

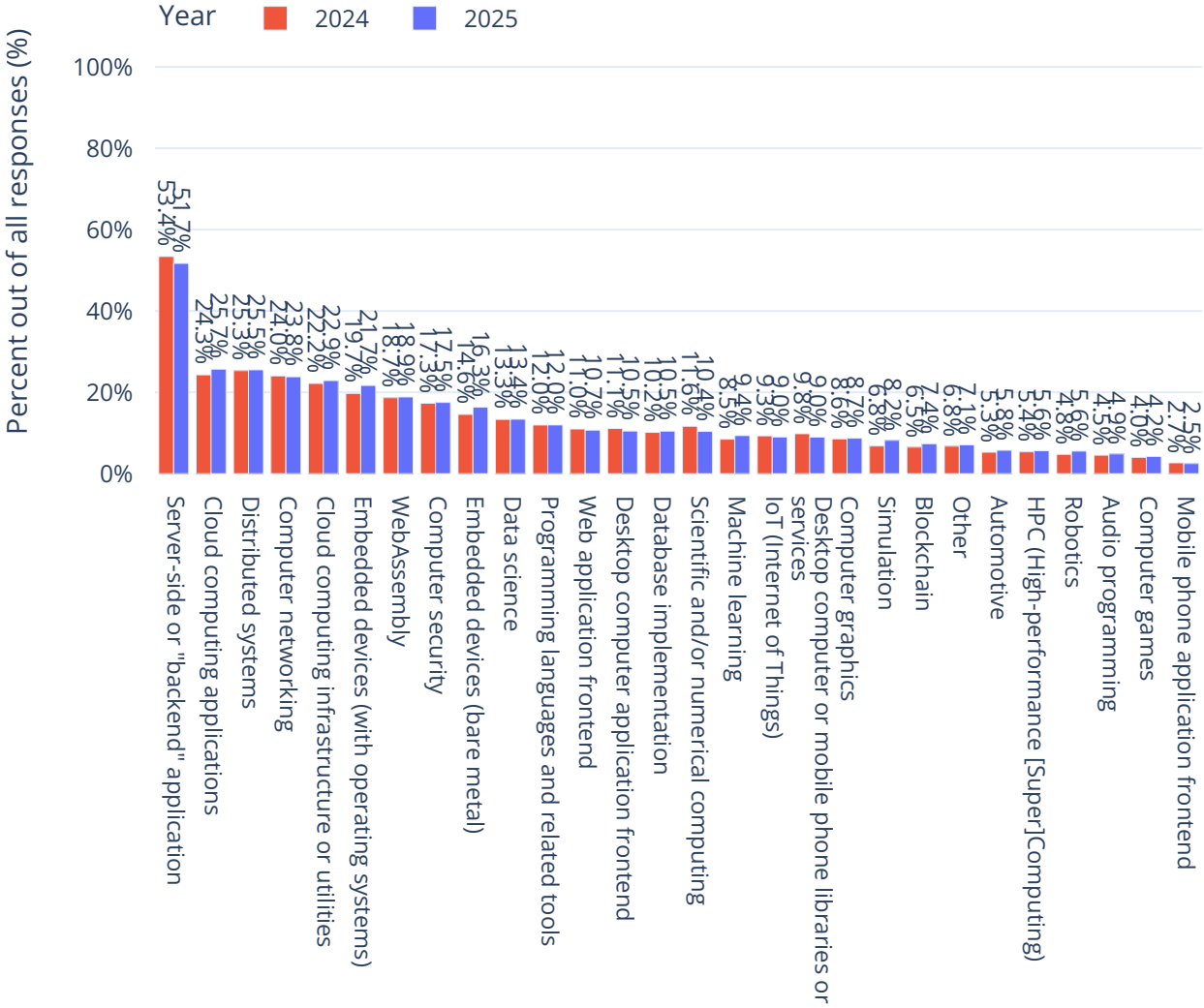
(total responses = 3281, multiple answers)



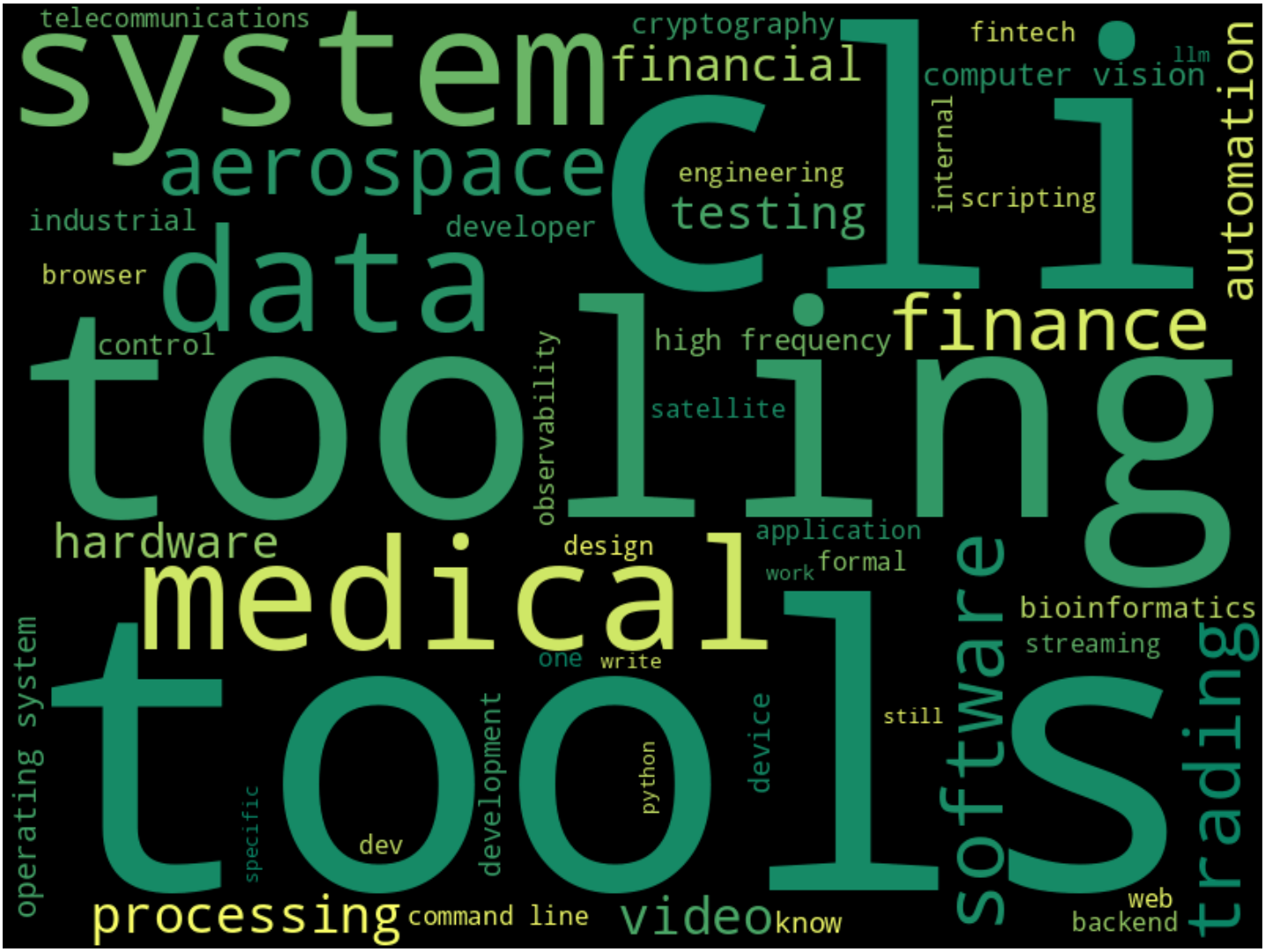

Which of the following statements are reasons why you use Rust at work?
(total responses = 3349, multiple answers)
Year
2022
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
82.1%
85.8%
87.2%
85.4%
82.9%
83.3%
84.5%
84.3%
67.6%
70.5%
74.8%
73.4%
72.6%
70.0%
71.2%
72.2%
37.4%
42.5%
46.9%
48.9%
13.5%
16.7%
21.2%
22.5%
12.8%
14.0%
16.4%
16.8%
8.5%
9.5%
9.8%
12.1%
It allows us to build relatively correct and bug free software
For its performance (i.e., speed, memory footprint, etc.) characteristics
Its security and safety properties are important to us
We find it enjoyable or fun to program in Rust
We need precise control over exactly how our software runs
We already know Rust so it's our default choice
We find it easy to prototype with
We must interact with existing Rust code

In what technology domain(s) is Rust used at your organisation?

(total responses = 3329, multiple answers)

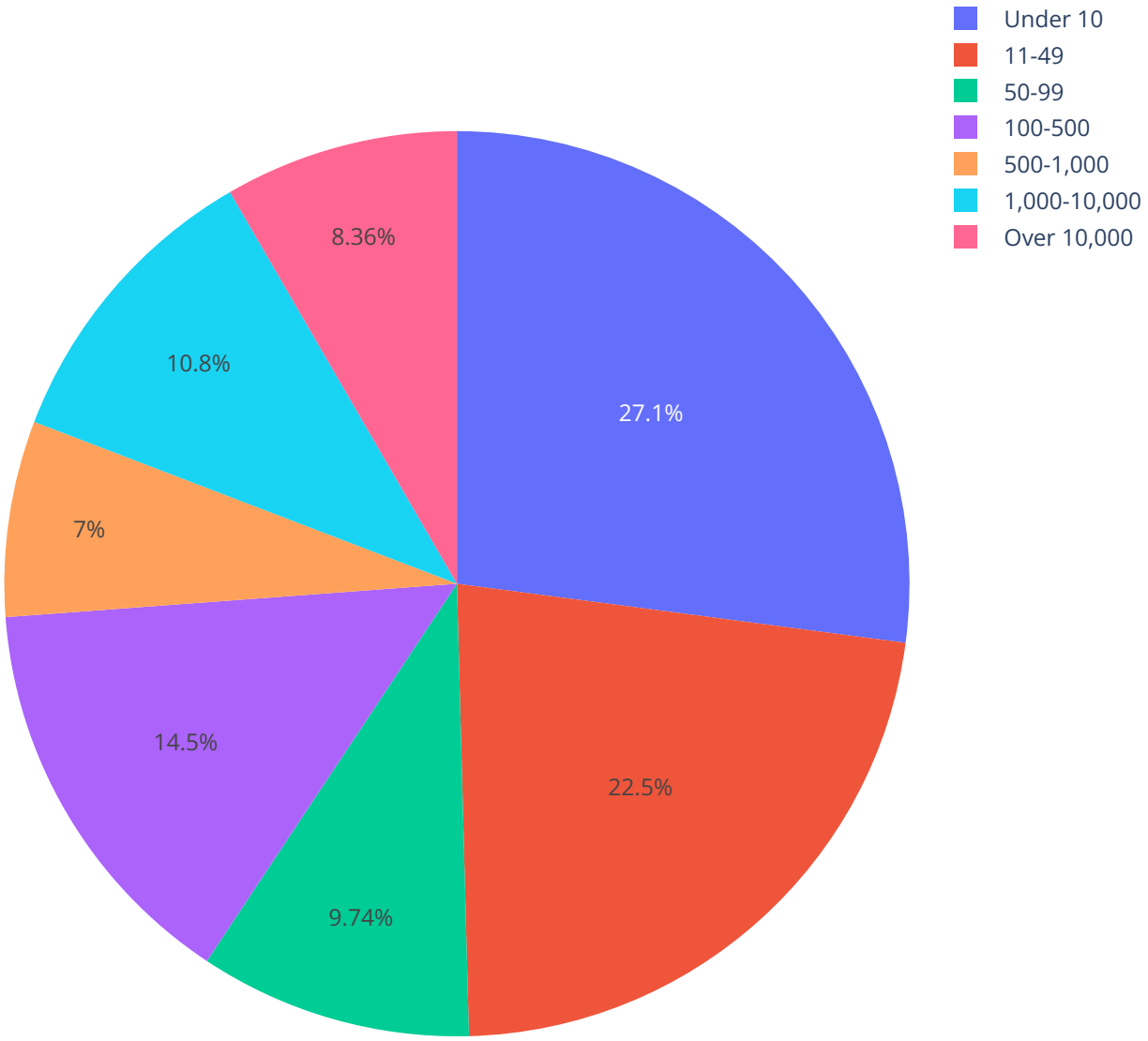


Wordcloud of open answers for the previous chart:



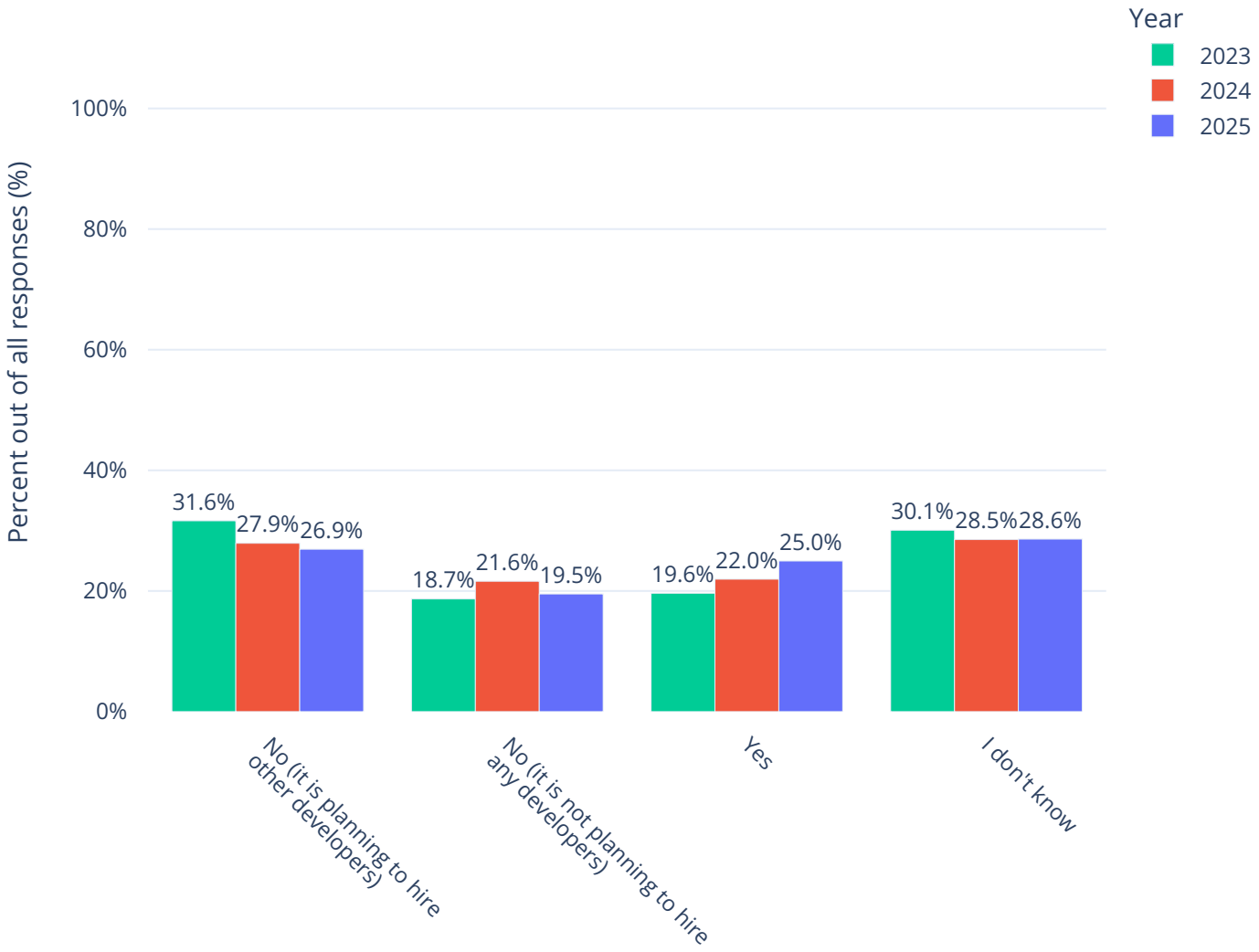
Approximately how many total developers does your organisation employ?

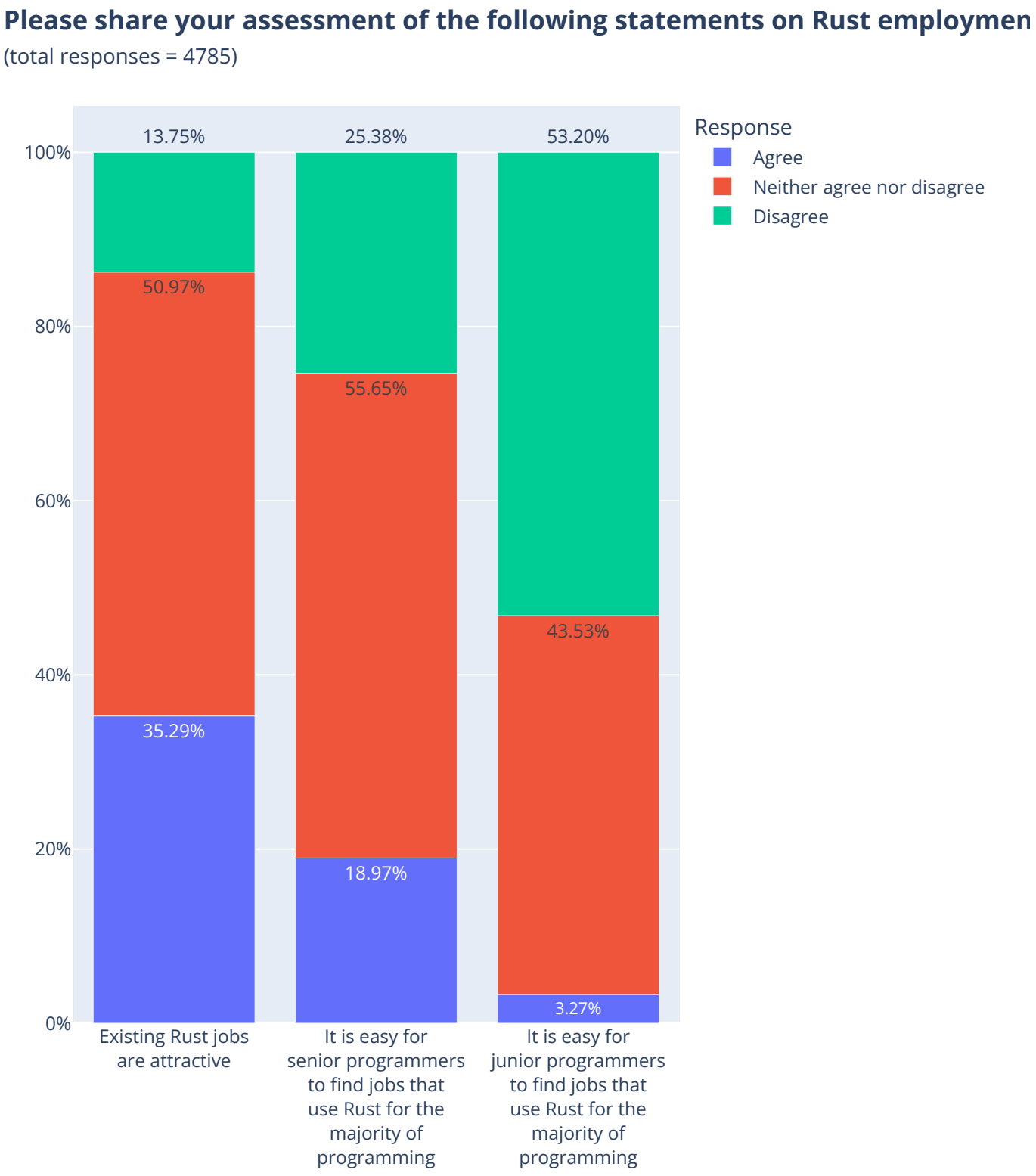
(total responses = 5073)

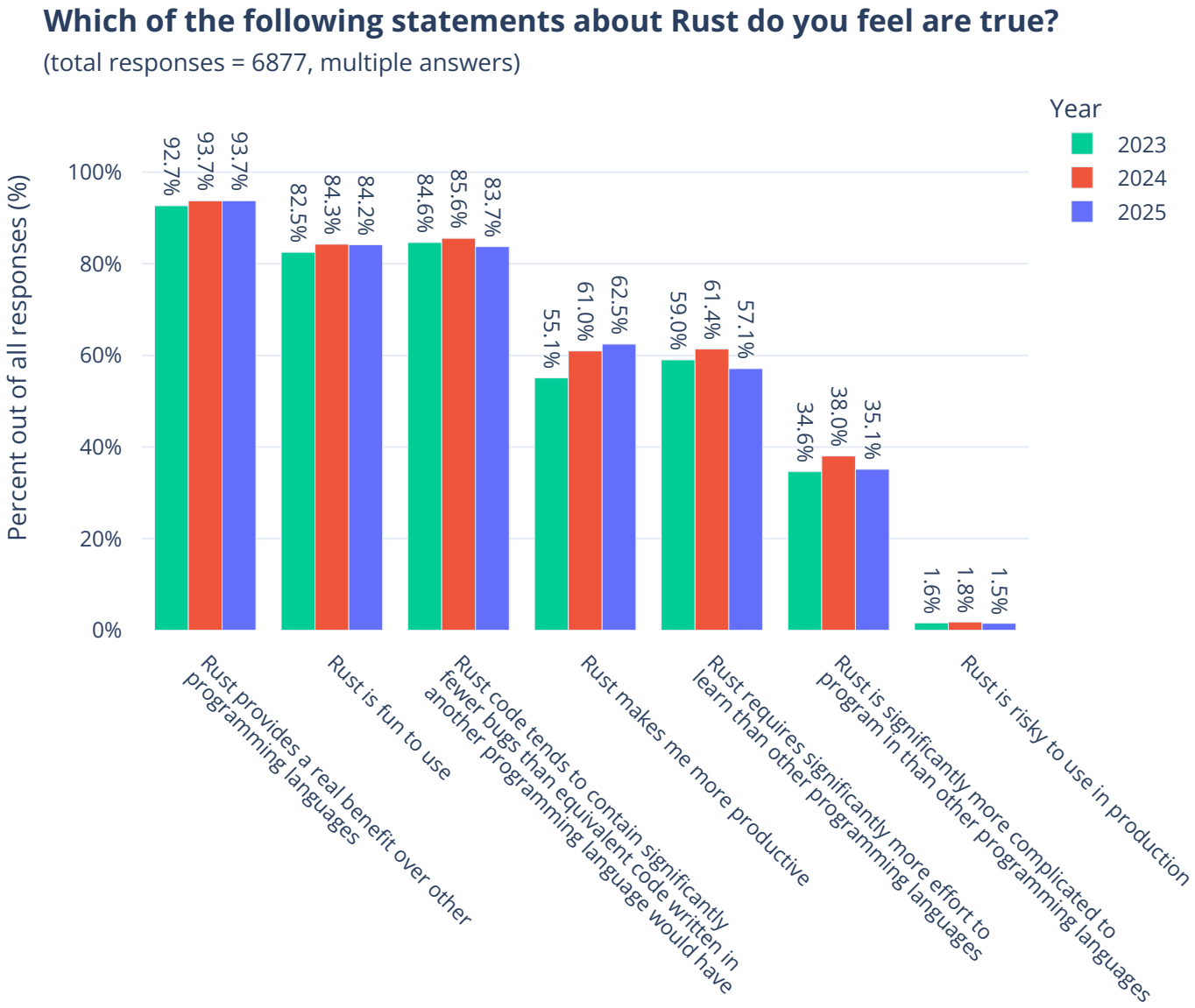


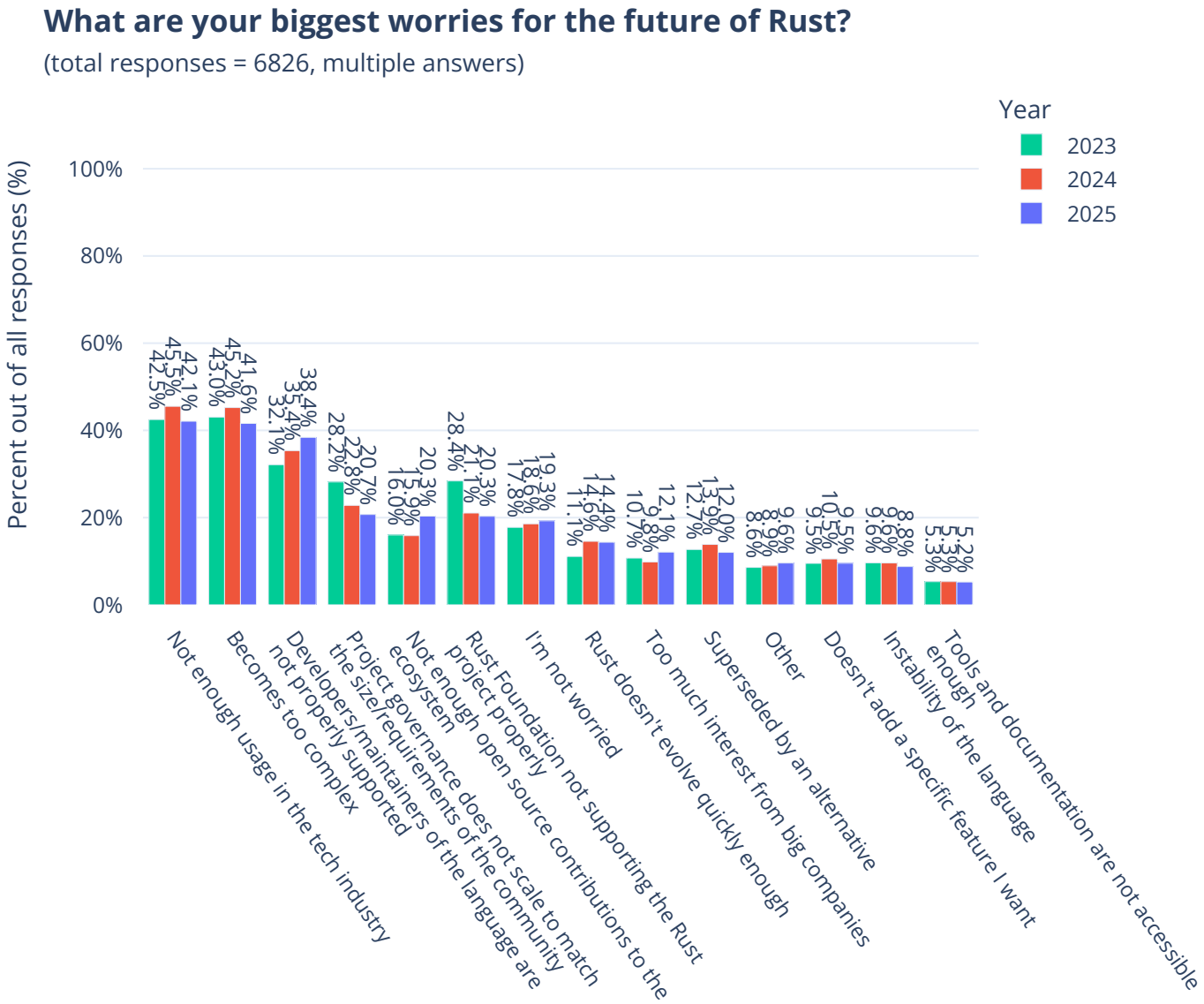
Is your organisation planning on hiring Rust developers in the next year?

(total responses = 5097, single answer)

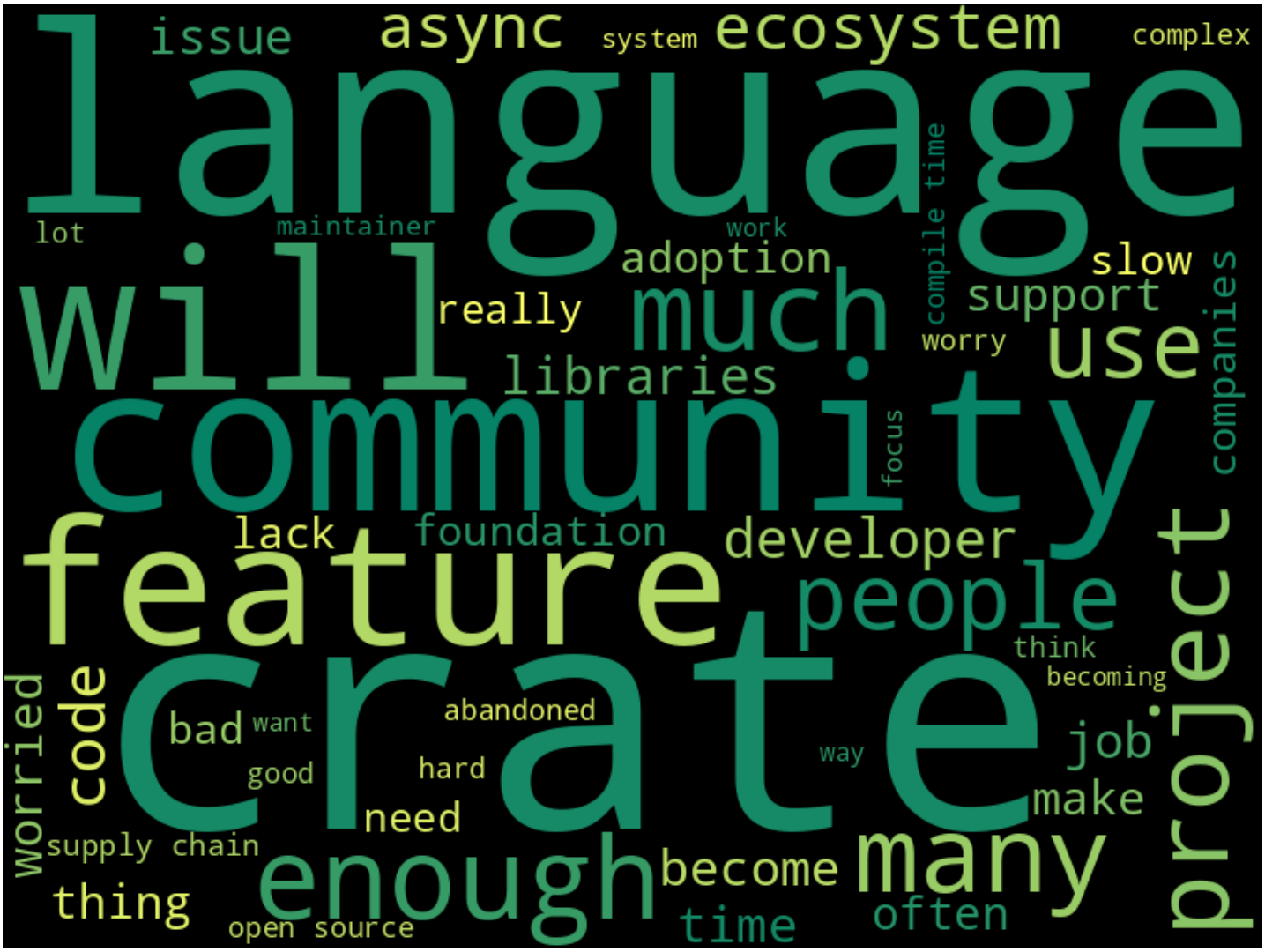



Please share your assessment of the following statements on Rust employmen
(total responses = 4785)
Response
Agree
Neither agree nor disagree
Disagree
100%
80%
60%
40%
20%
0%
13.75%
50.97%
35.29%
25.38%
55.65%
18.97%
53.20%
43.53%
3.27%
Existing Rust jobs are attractive
It is easy for senior programmers to find jobs that use Rust for the majority of programming
It is easy for junior programmers to find jobs that use Rust for the majority of programming


Which of the following statements about Rust do you feel are true?
(total responses = 6877, multiple answers)
Year
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
92.7%
93.7%
93.7%
82.5%
84.3%
84.2%
84.6%
85.6%
83.7%
55.1%
61.0%
62.5%
59.0%
61.4%
57.1%
34.6%
38.0%
35.1%
1.6%
1.8%
1.5%
Rust provides a real benefit over other programming languages
Rust is fun to use
Rust code tends to contain significantly fewer bugs than equivalent code written in another programming language would have
Rust makes me more productive
Rust requires significantly more effort to learn than other programming languages
Rust is significantly more complicated to program in than other programming languages
Rust is risky to use in production

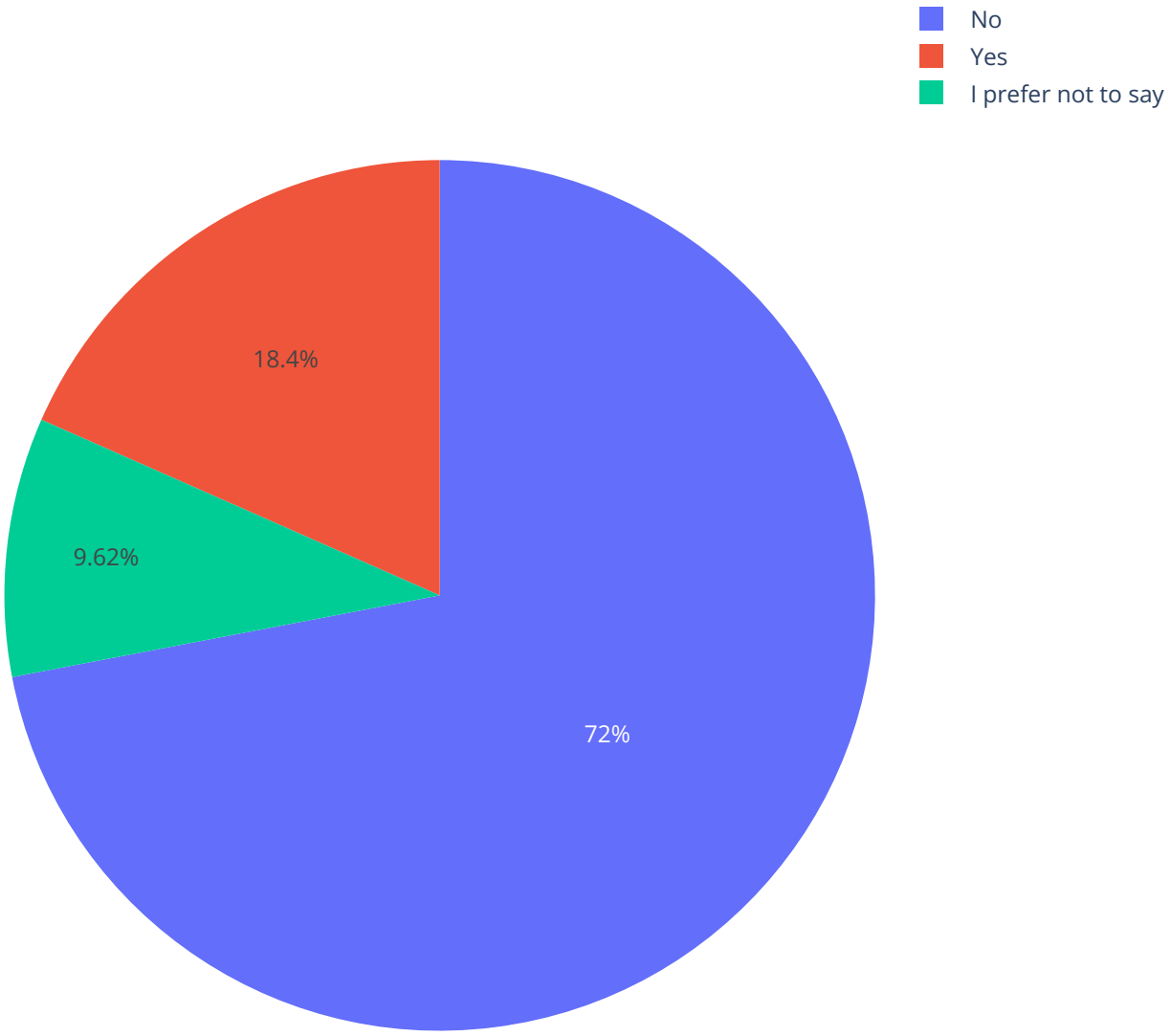

What are your biggest worries for the future of Rust?
(total responses = 6826, multiple answers)
Year
2023
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
42.5%
45.5%
42.1%
43.0%
45.2%
41.6%
32.1%
35.4%
38.4%
28.2%
22.8%
20.7%
16.0%
15.9%
20.3%
28.4%
21.1%
20.3%
17.8%
18.6%
19.3%
11.1%
14.6%
14.4%
10.7%
9.8%
12.1%
12.7%
13.9%
12.0%
8.6%
8.9%
9.6%
9.5%
10.5%
9.5%
9.6%
9.6%
8.8%
5.3%
5.3%
5.2%
Not enough usage in the tech industry
Becomes too complex
Developers/maintainers of the language are not properly supported
Project governance does not scale to match the size/requirements of the community
Not enough open source contributions to the ecosystem
Rust Foundation not supporting the Rust project properly
I'm not worried
Rust doesn't evolve quickly enough
Too much interest from big companies
Superseded by an alternative
Other
Doesn't add a specific feature I want
Instability of the language
Tools and documentation are not accessible enough

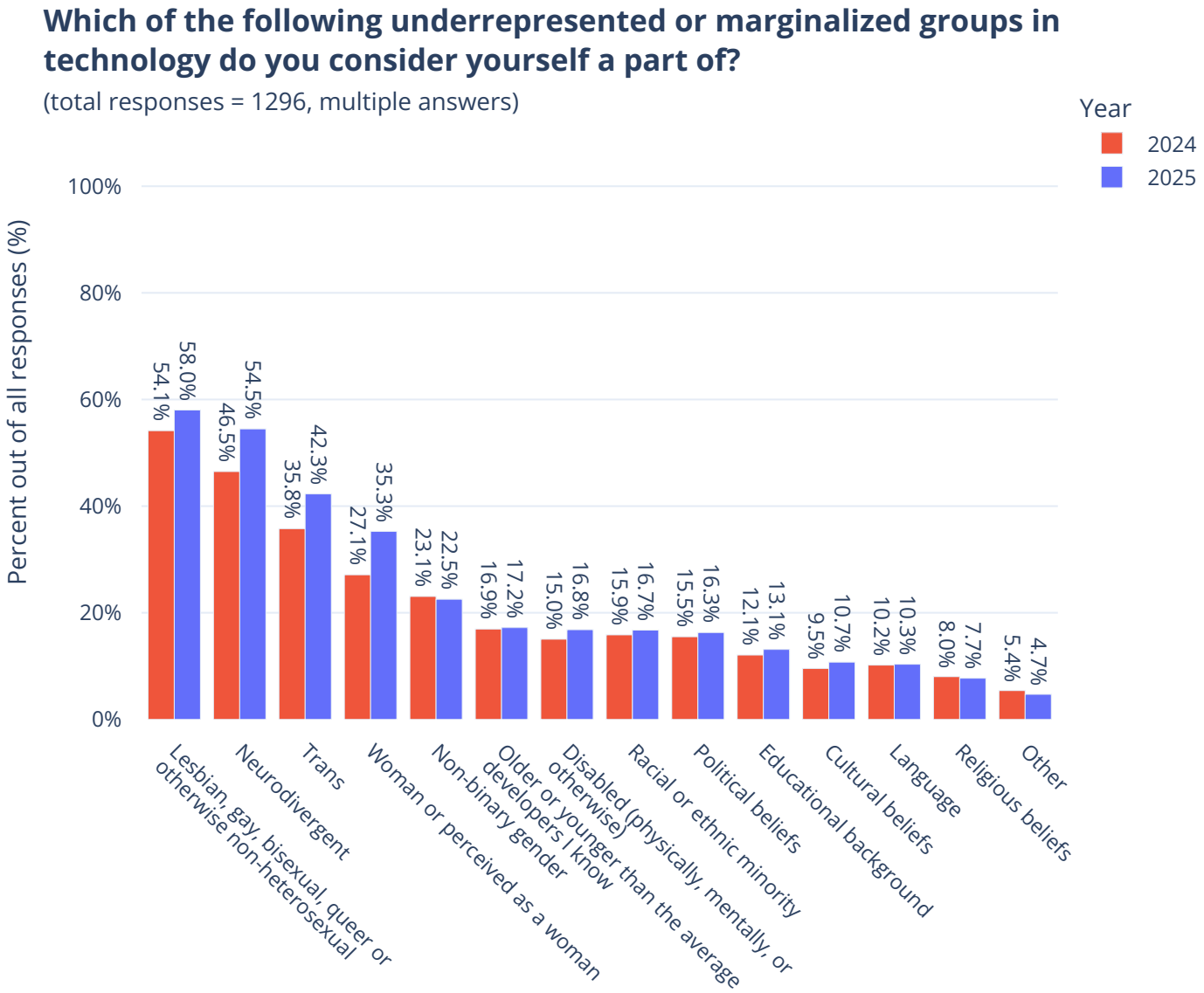
Wordcloud of open answers for the previous chart:



Do you consider yourself a member of a group that is underrepresented or marginalized in technology?

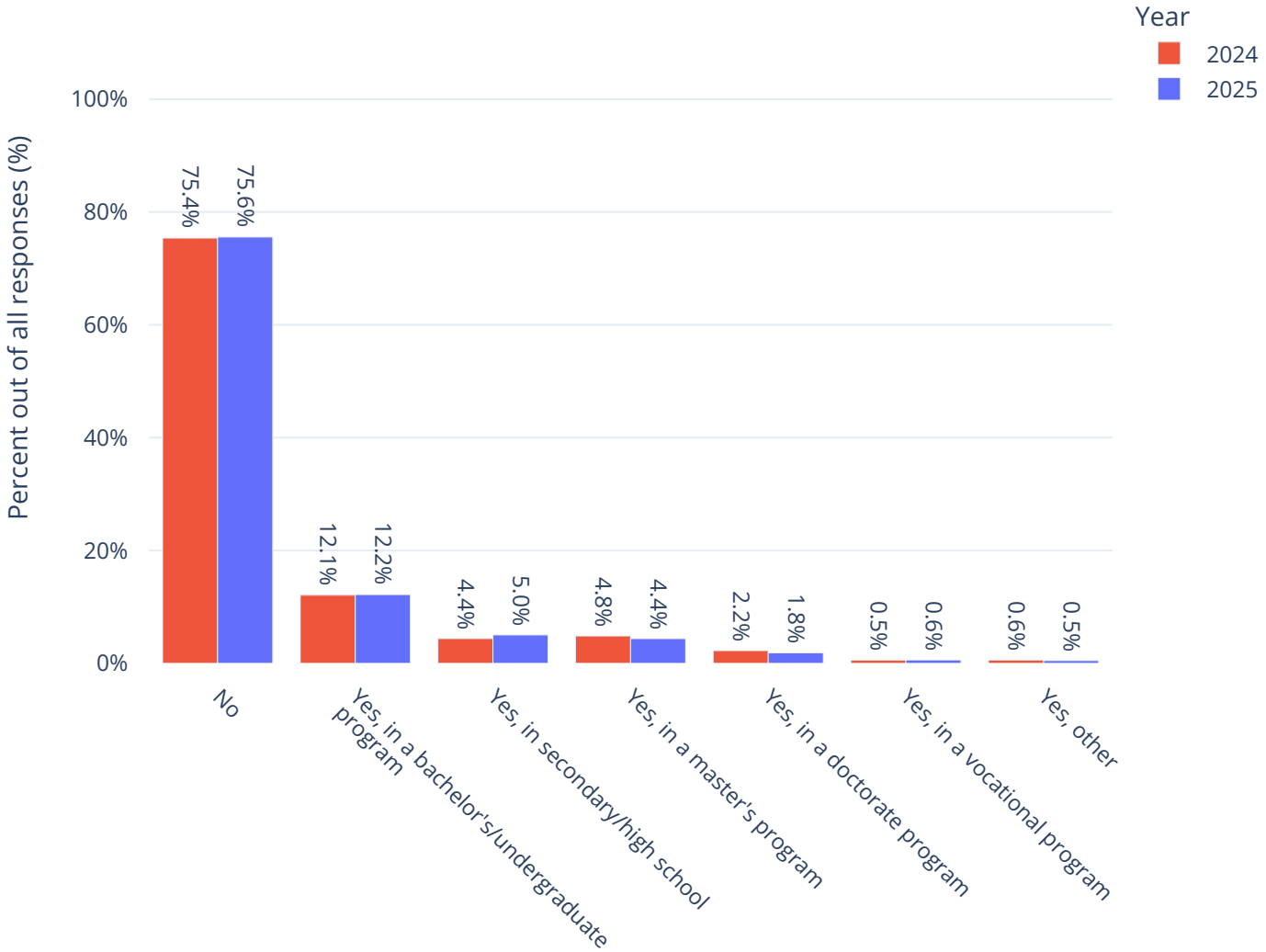
(total responses = 7102)




Which of the following underrepresented or marginalized groups in technology do you consider yourself a part of?
(total responses = 1296, multiple answers)
Year
2024
2025
Percent out of all responses (%)
100%
80%
60%
40%
20%
0%
54.1%
58.0%
46.5%
54.5%
35.8%
42.3%
27.1%
35.3%
23.1%
22.5%
16.9%
17.2%
15.0%
16.8%
15.9%
16.7%
15.5%
16.3%
12.1%
13.1%
9.5%
10.7%
10.2%
10.3%
8.0%
7.7%
5.4%
4.7%
Lesbian, gay, bisexual, queer or otherwise non-heterosexual
Neurodivergent
Trans
Woman or perceived as a woman
Non-binary gender
Older or younger than the average developers I know
Disabled (physically, mentally, or otherwise)
Racial or ethnic minority
Political beliefs
Educational background
Cultural beliefs
Language
Religious beliefs
Other

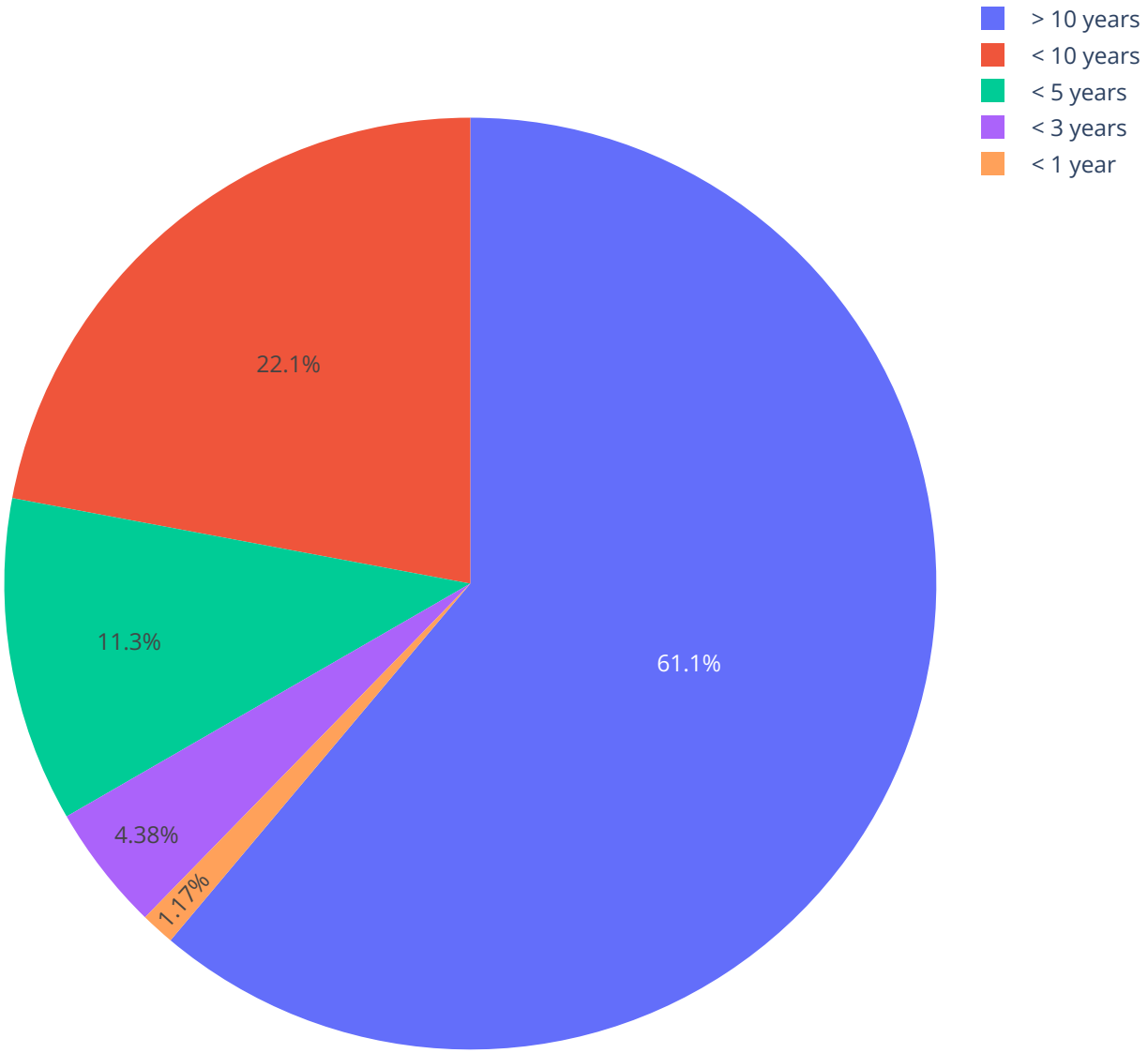
Are you a full- or part-time student?

(total responses = 7127, single answer)



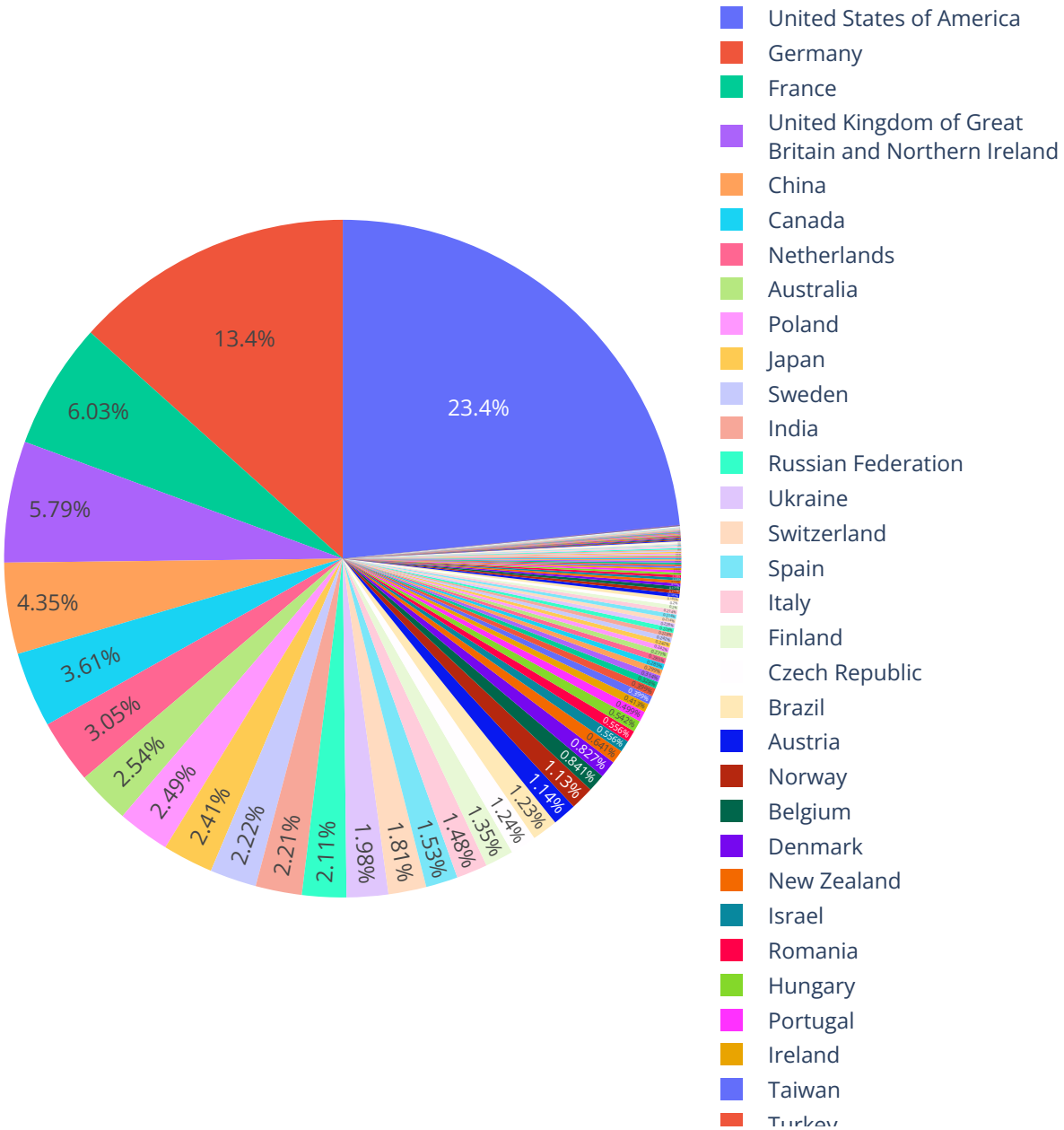
How long have you been programming (in any language, for any reason)?

(total responses = 7153)



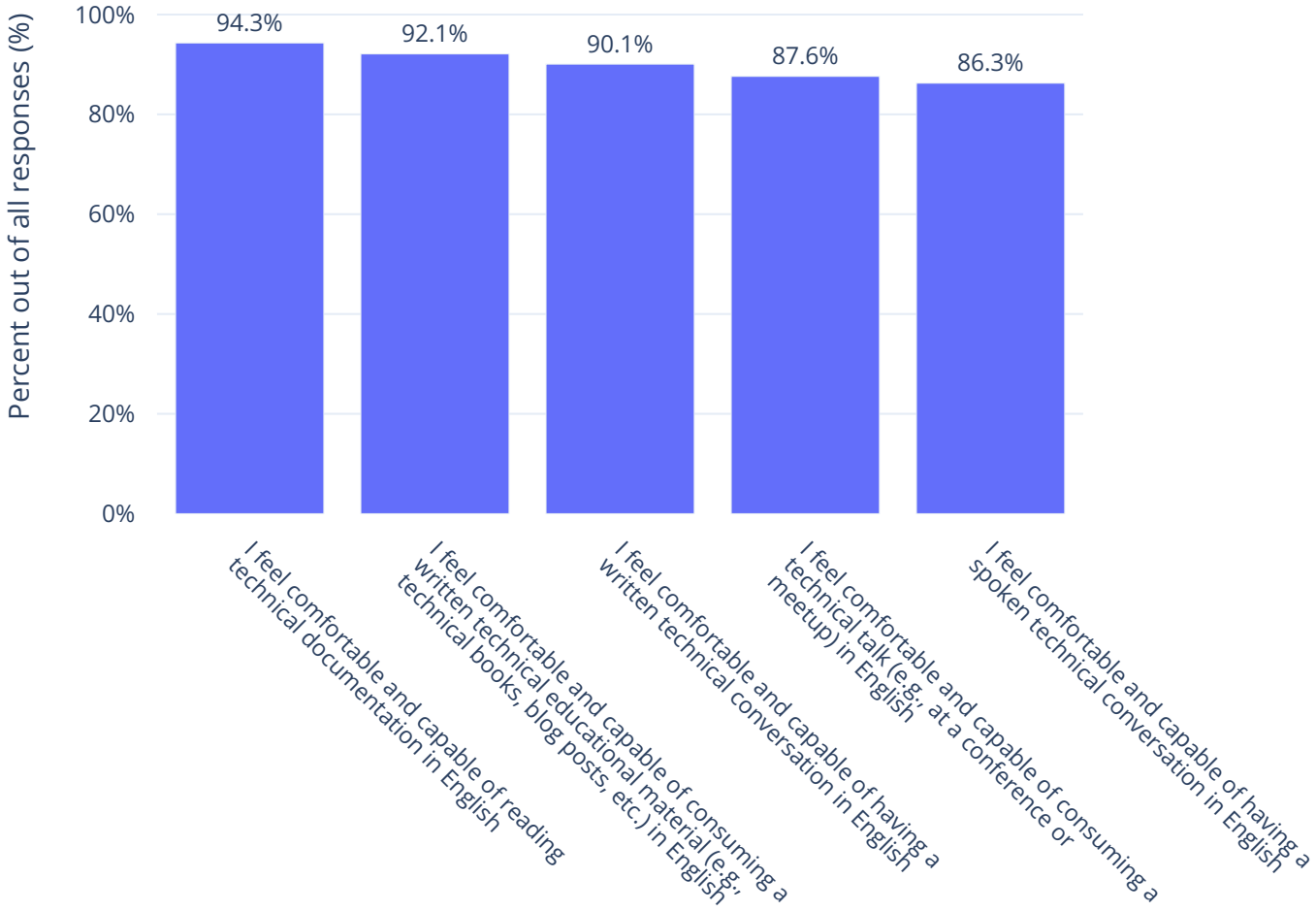
Where do you live?

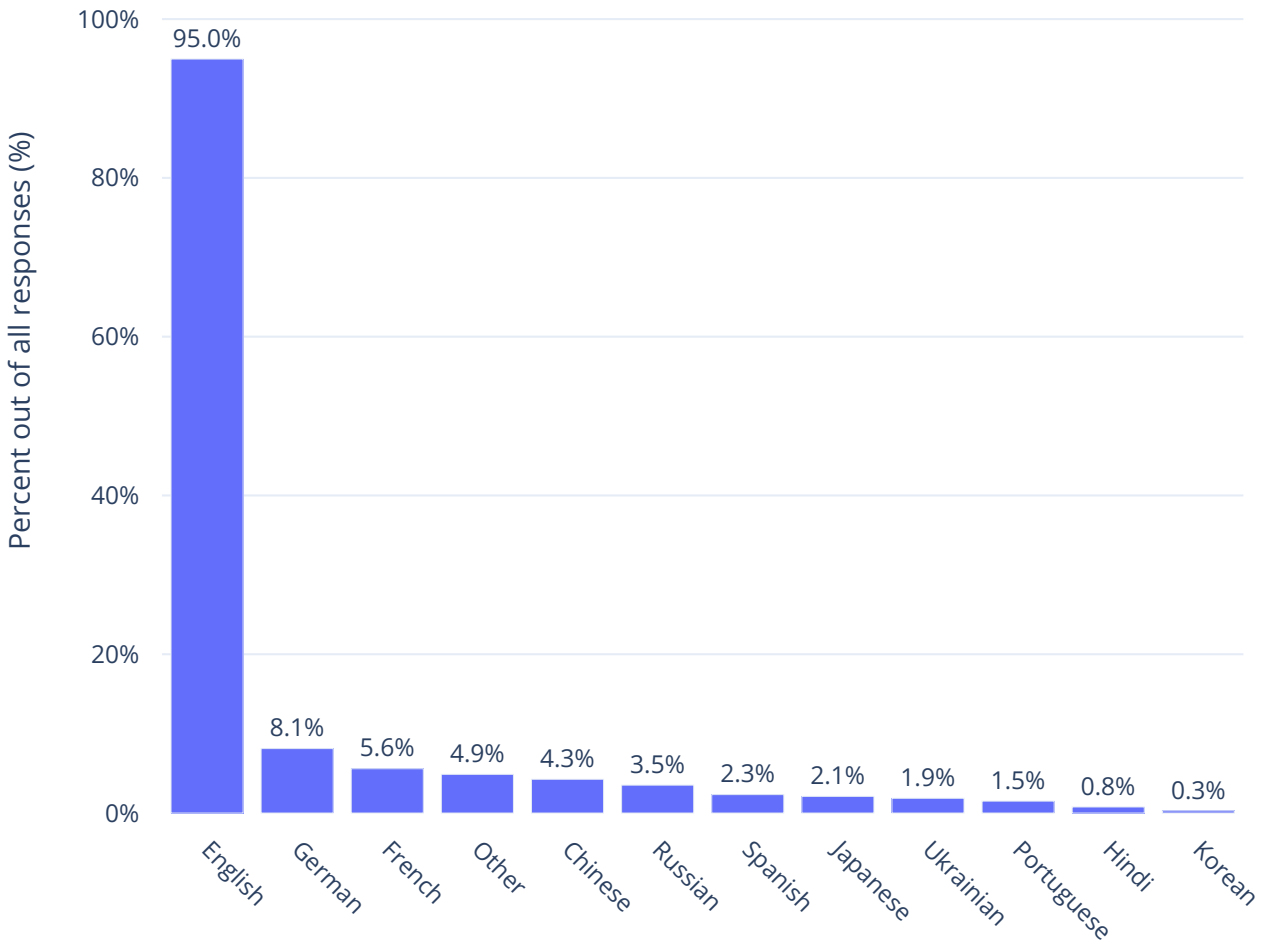
(total responses = 7015)



In what ways are you comfortable communicating about technical topics in English?

(total responses = 7048, multiple answers)



In which language(s) are you most comfortable when consuming existing technical content (e.g. blog posts, documentation, etc.)?

(total responses = 7058, multiple answers)